

EEE

321

Bashundhara
Exercise Book
Write Your Future

Reem

1802071

Reference Books

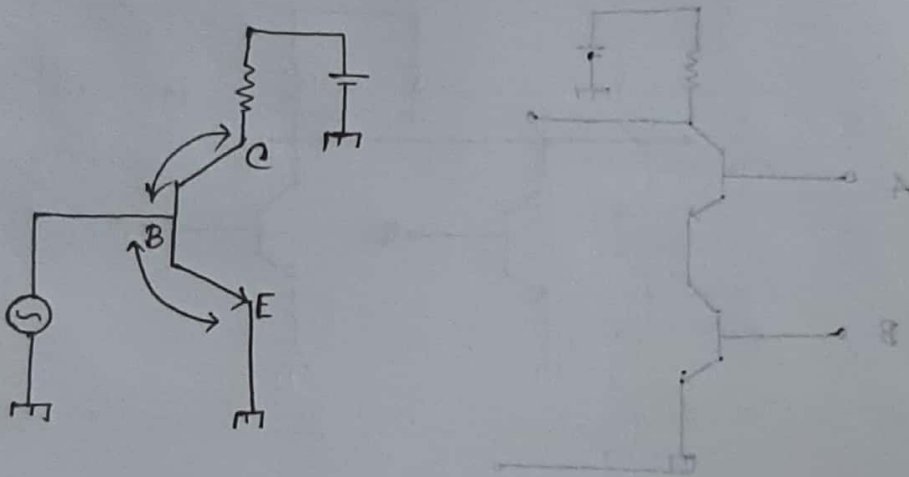
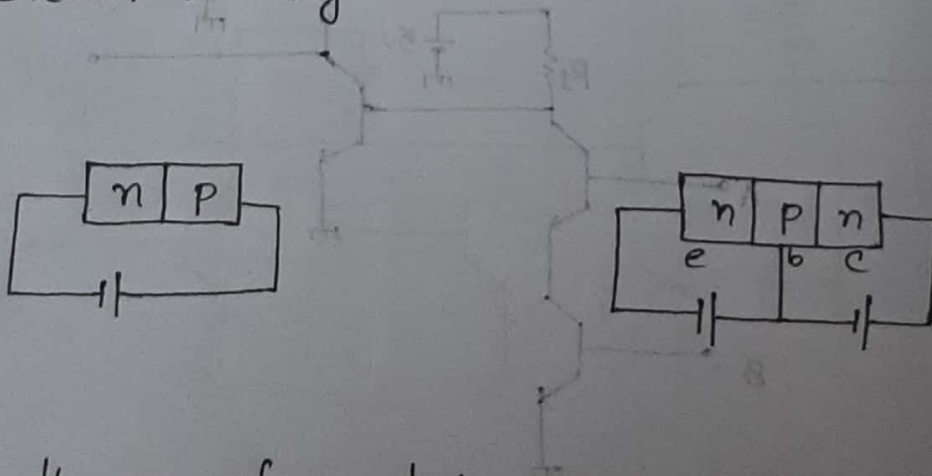
1. Electronic Devices and Circuit Theory '

Robert L. Boylestad

2. Modern Digital Electronics

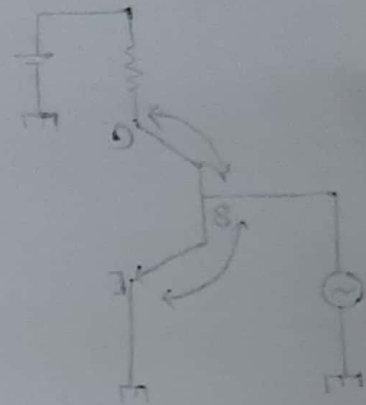
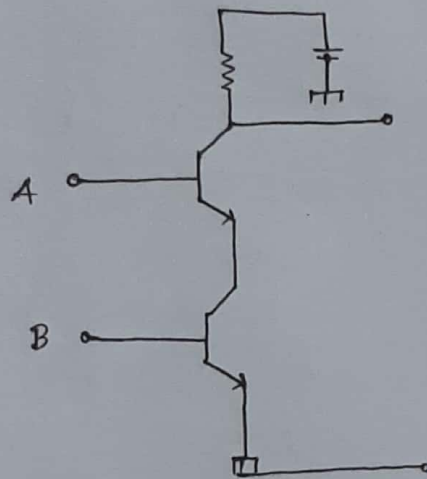
RP Jain (2nd Edition)

Chapter 04

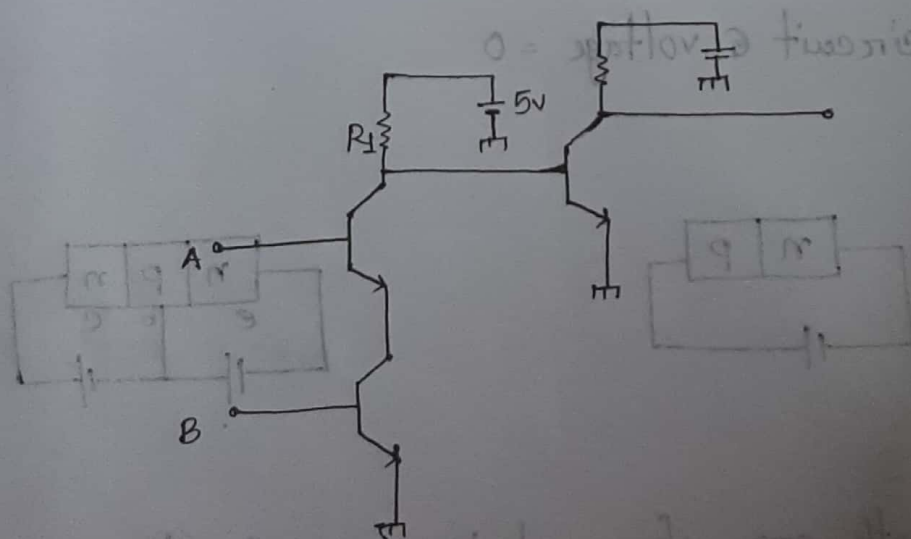
RTL $\rightarrow$ Register Transistor LogicOpen circuit V voltage = input voltageShort circuit V voltage = 0When both are forward bias $\rightarrow$ saturationWhen " " reverse " $\rightarrow$ cut off region

Logic gates are made with diode and transistor.

NAND Gate:



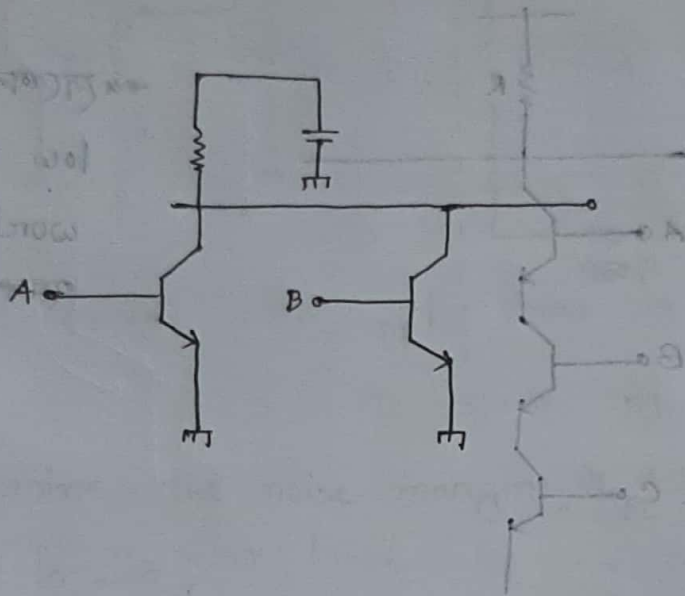
AND Gate:



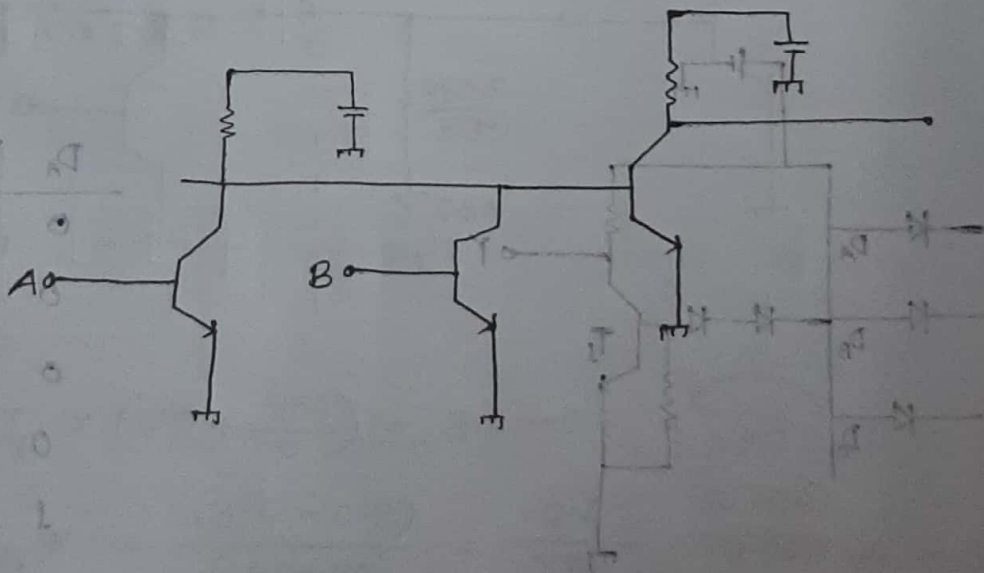
When both are forward bias $\rightarrow$ saturation
When $\rightarrow$ cut off region

* Design RTL NOR gate and explain working principle.

NOR Gate:



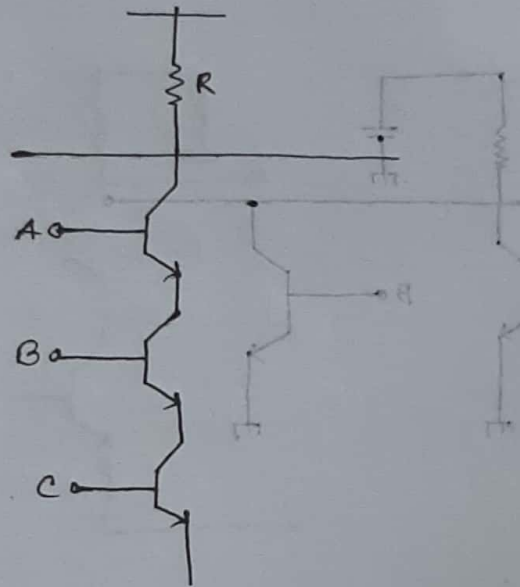
OR Gate:



** Diode, transistor किताव बजावले? कोभास कोभास use रय?

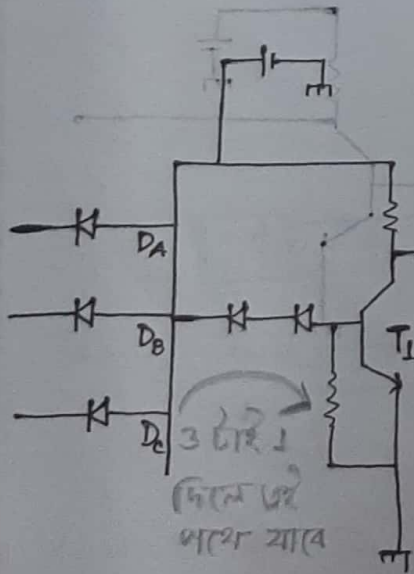
Diode, Transistor, Noise margin (noise tolerance level), Fanout, Propagation Delay.

DTL → Diode Transistor Logic



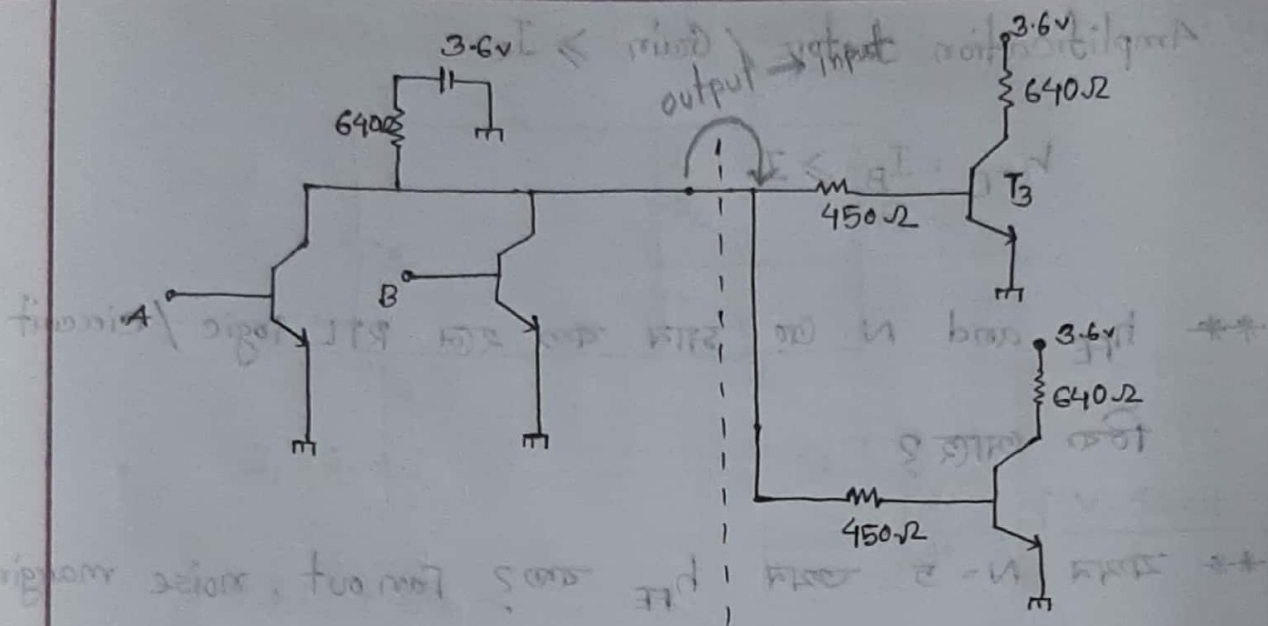
*** (सबसे) Diode को 0 or low voltage मिल जाय
working principle समझें ***

Three Input NAND gate

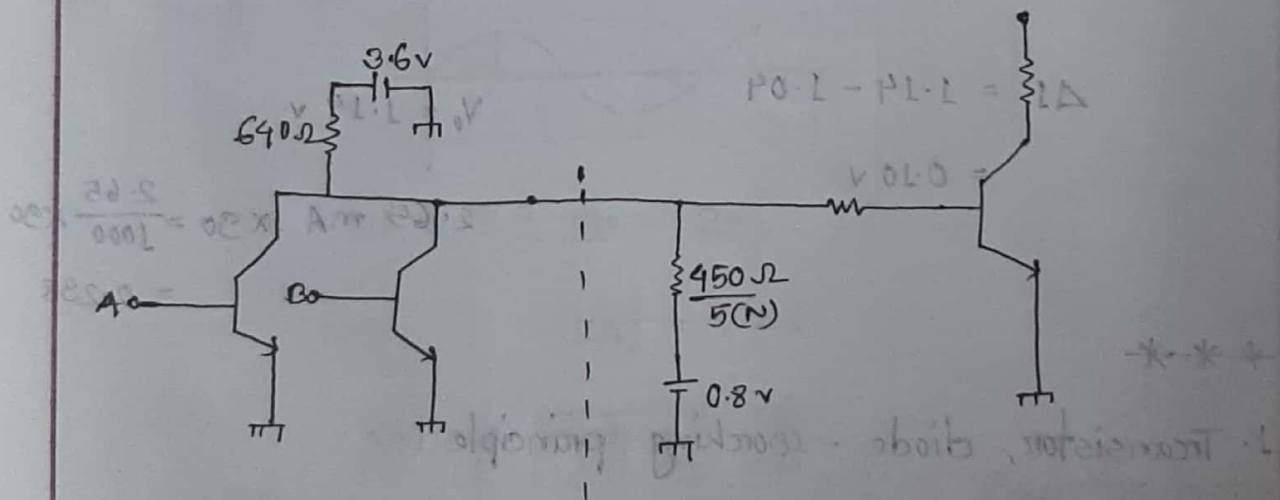


D_A	D_B	D_C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

input 1 सबसे पहले 0 रखें
output 1 रखें



** Determine the noise margin, h_{FE} of the following circuit while $m = 5$.



$$I_B \times \left(640 + \frac{450}{5}\right) = 3.6 - 0.8$$

$$\Rightarrow I_B = \frac{(3.6 - 0.8)}{(640 + 90)} = \frac{2.8}{730} = 0.0038 \text{ A} = 3.8 \text{ mA}$$

$$-3.6 + 640 \times I_C + 0.2 = 0$$

$$\Rightarrow I_C = \frac{3.6 - 0.2}{640} = 0.0053 \text{ A} = 5.3 \text{ mA}$$

Amplification factor / Gain $\geq I_c$

$$h_{FE} \cdot I_B \geq I_c$$

** h_{FE} and N এর জান বন্ধ হলে RTL logic / circuit চিক আছে?

** যখন $N=5$ তখন h_{FE} বন্ধ? Fan out, noise margin বন্ধ

- Noise margin বের করতে হবে।

$$\Delta 0 = 0.5 - 0.2$$

$$= 0.3 \rightarrow \text{zero level noise margin}$$

$$\Delta 1 = 1.14 - 1.04$$

$$= 0.10 \text{ V}$$

$$V_0 = 1.14 \text{ V}$$

$$2.65 \text{ mA} \times 90 = \frac{2.65}{1000} \times 90 = 0.238$$

1. Transistor, diode - working principle

2. V-I curve

3. Active, cut off, saturation region

4. FET / MOSFET

5. KVL, KCL

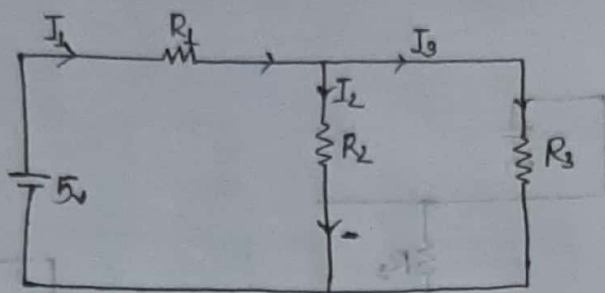
6. RLC Circuit

Trans + reg $\rightarrow$ Low to high reg

$\sum \text{incoming current} = \sum \text{outgoing current}$

Charge vs Current $\rightarrow$

বিভিন্ন উপাদান charge and current



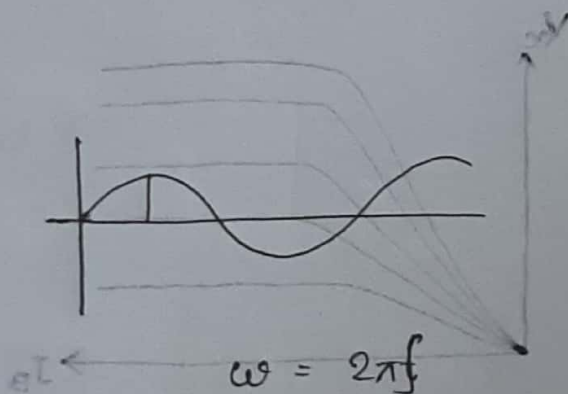
$$I_1 = I_2 + I_3$$

$$-5 + I_1 R_1 + I_2 R_2 = 0$$

$$\Rightarrow I_1 R_1 + I_2 R_2 = 5$$

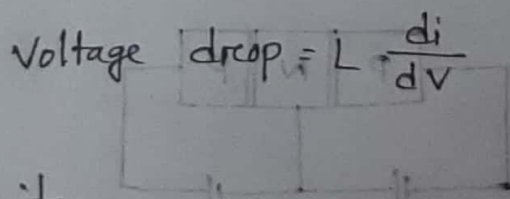
$$\int \frac{di}{dt} = \int \frac{V_m \sin \omega t}{L} = \frac{V_m}{\omega L} \cos \omega t$$

$$V = V_m \sin \omega t$$



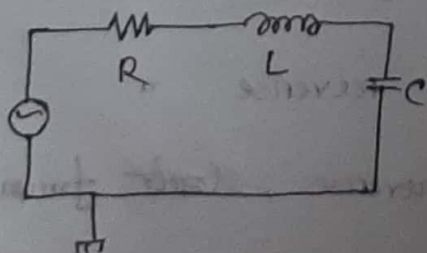
$$\omega = 2\pi f$$

$$2\pi f = 200t$$

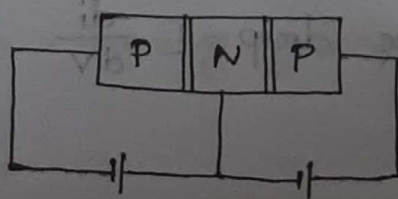
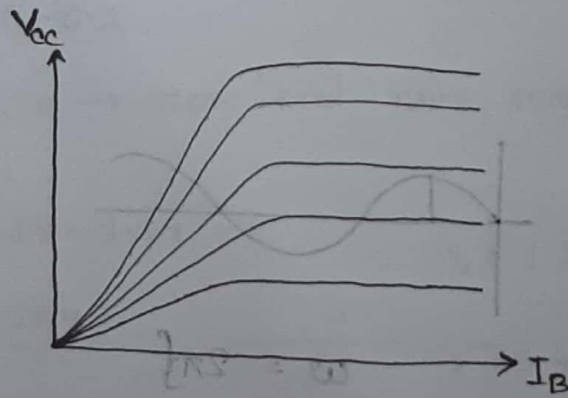
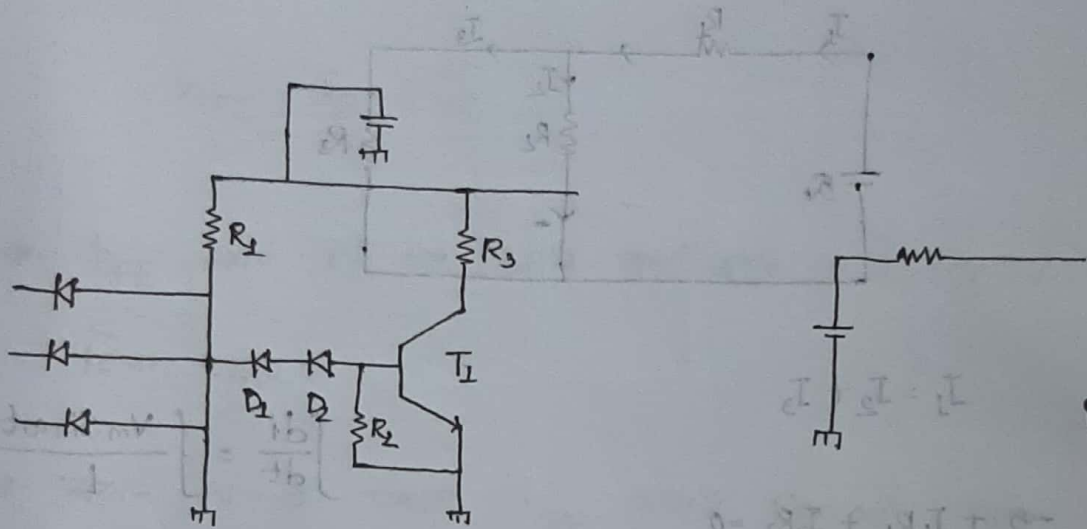


$$\text{Capacitor voltage drop} = \frac{1}{C} \int i dt$$

RLC circuit



$$\sqrt{R^2 + (X_L - X_C)^2}$$

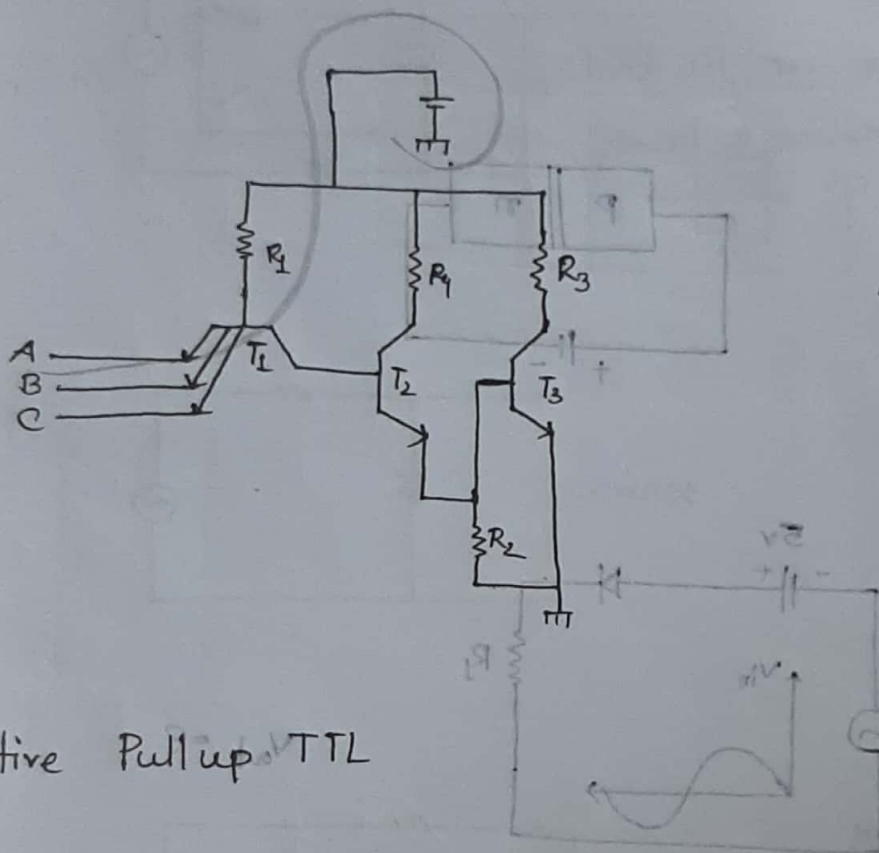
DTL $\rightarrow$ Diode Transistor Logic

Saturation $\rightarrow$ यहाँ β का forward biased

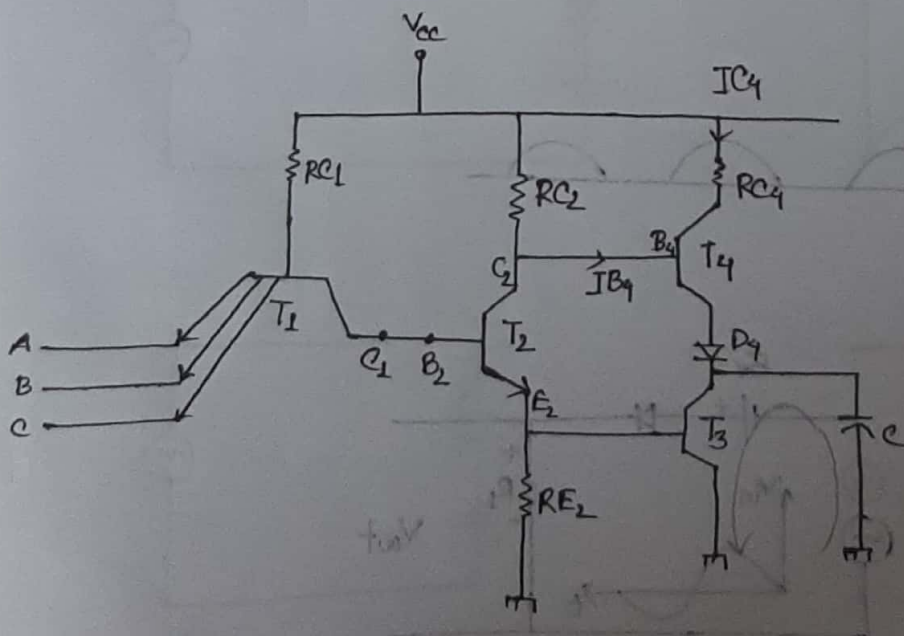
Cut off region $\rightarrow$ " β " reverse "

Active region $\rightarrow$ यहाँ reverse, यहाँ forward

TTL → Transistor Transistor Logic



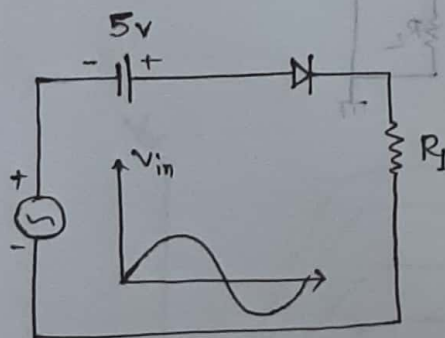
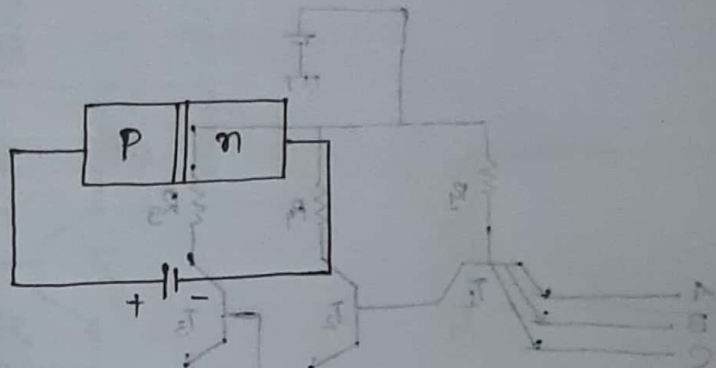
Active Pullup TTL



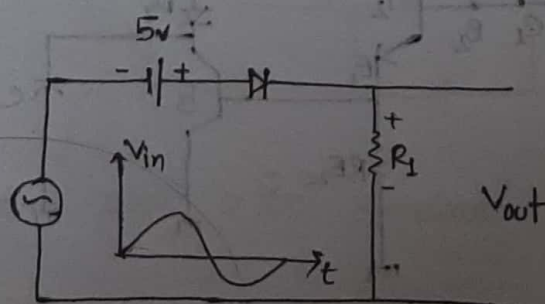
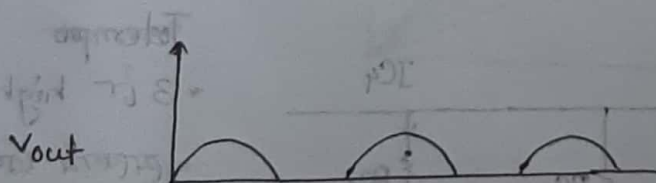
Totem Pole Output

Chapter 02 (Boylestad)

Clippers



$$V_{out} = R_1 I_{out}$$



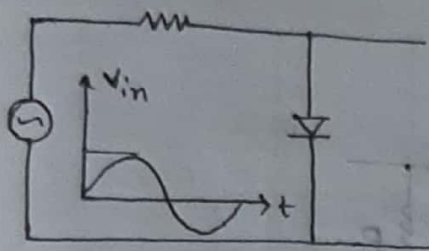
$$-V_{in} - 5V + V_{out} = 0$$

$$\Rightarrow V_{out} = V_{in} + 5V$$

$$-V_{in} + 5 + V_{out} = 0$$

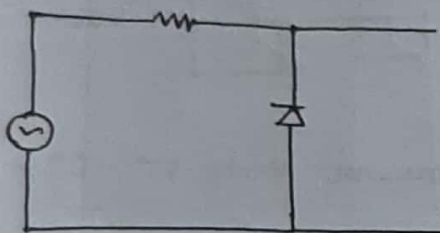
$$\Rightarrow V_{out} = V_{in} - 5$$

AC, DC current
 use KVL, KCL
 apply

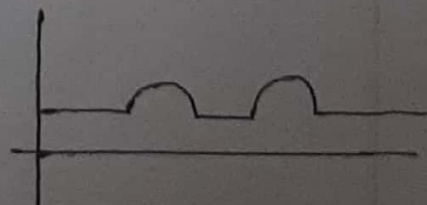
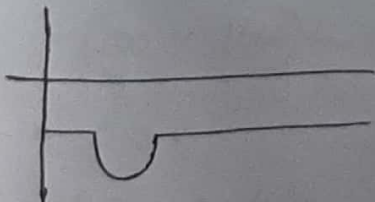
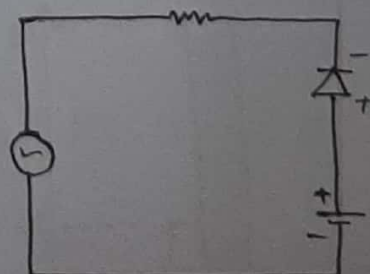
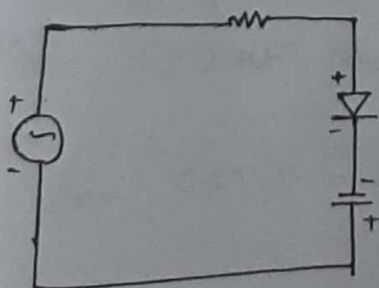
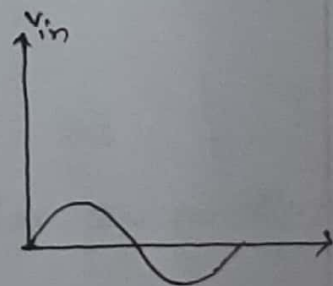
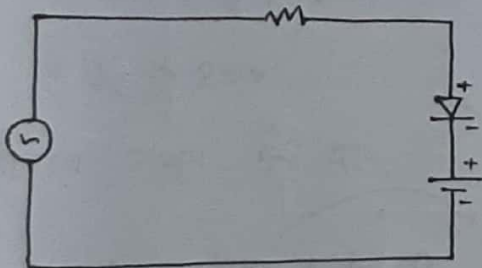


short हो कर वाह करे.

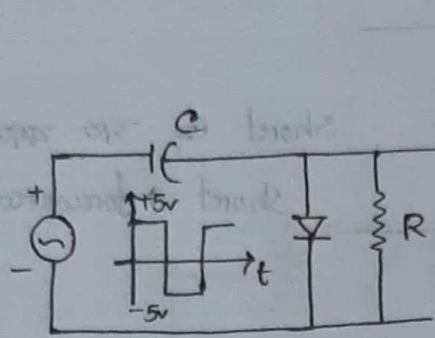
short $\rightarrow$ forward



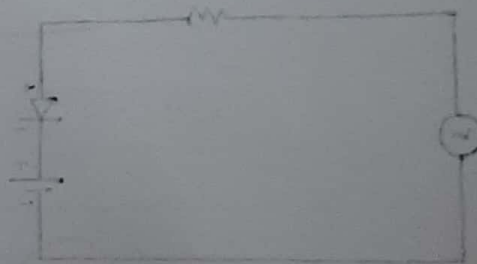
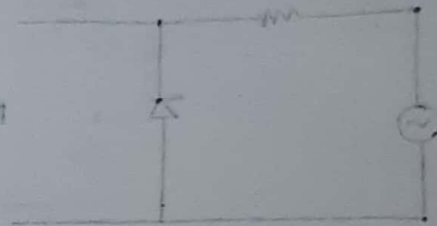
reverse



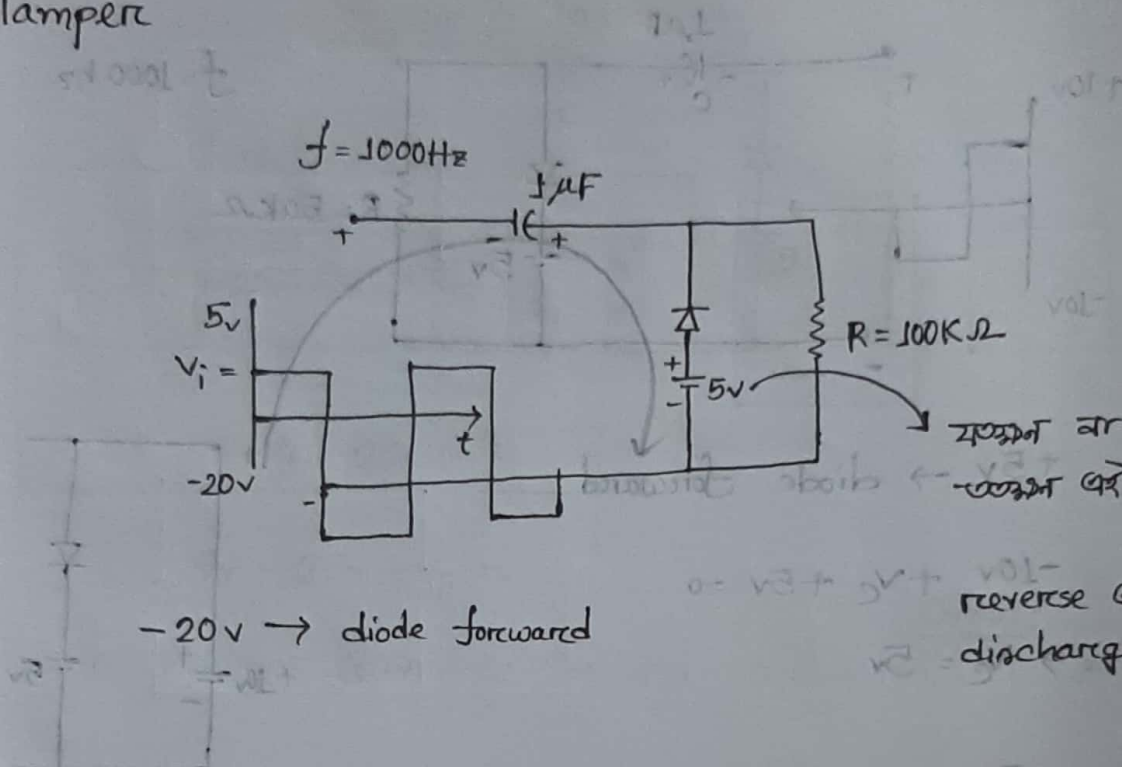
Clamper



Clamper for viva



Clamper



$$R = 100 K\Omega, C = 1 \mu F, f = 1000 Hz$$

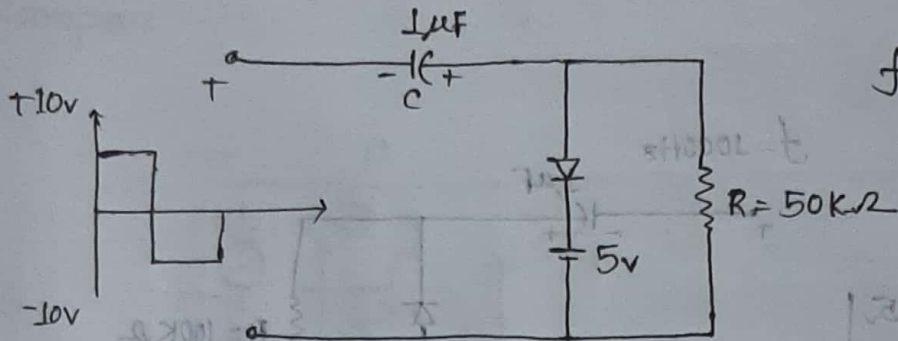
$$T = \frac{1}{f} = \frac{1}{1000} = 0.001 s = 1 ms$$

$$\tau = R \times C$$

$$= (100 \times 1000) \Omega \times 10^{-6} F$$

$$= 0.1 s$$

$$5\tau = 0.5 s > 1 ms. \text{ Since } 5\tau > T; \text{ the circuit is OK.}$$



+5V → diode forward

$$-10V + V_C + 5V = 0$$

$$\Rightarrow V_C = 5V$$

$$-5V + V_R - 5V = 0$$

$$\Rightarrow V_R = 10V$$

$$T = \frac{1}{f} = \frac{1}{1000} = 0.001 \text{ s} = 1 \text{ ms}$$

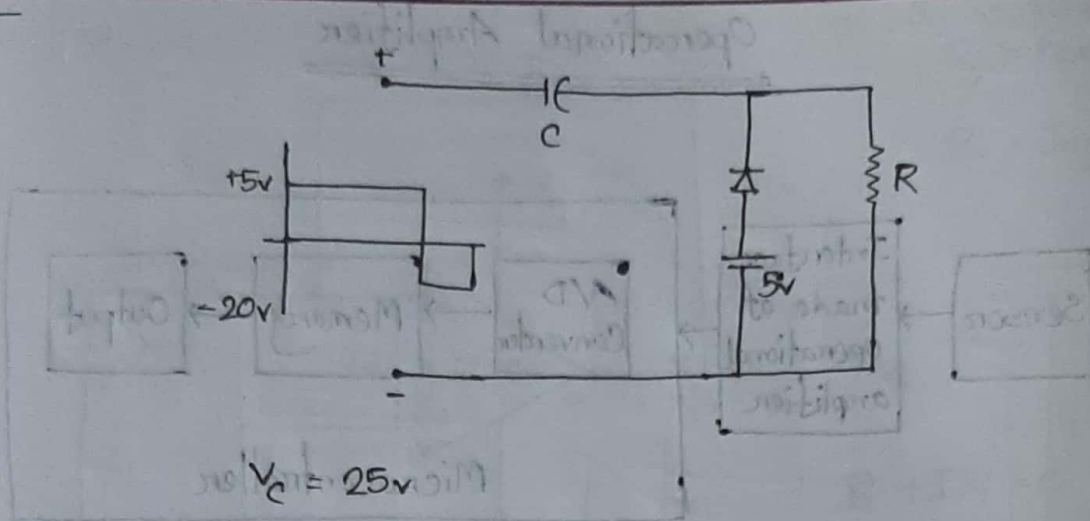
$$\tau = R \times C$$

$$= 50 \times 1000 \times 10^{-6}$$

$$= 0.05 \text{ s}$$

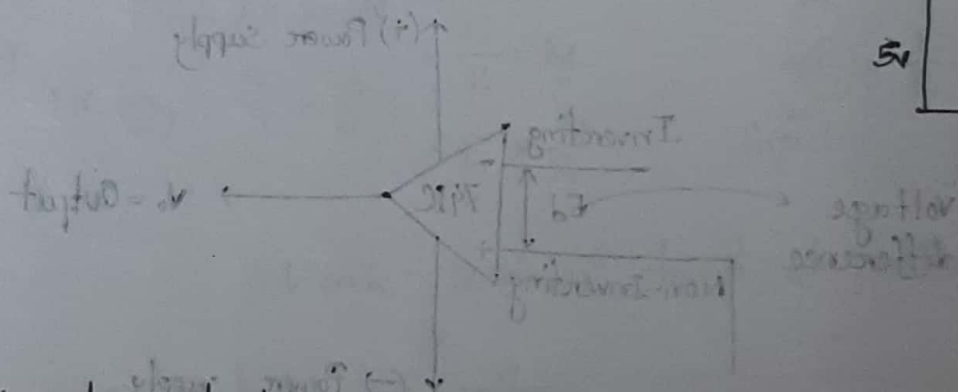
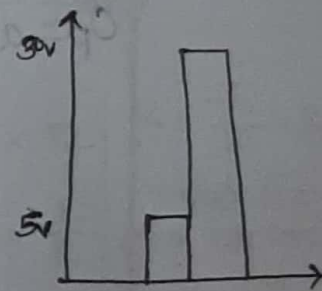
$$5\tau = 0.25 \text{ s} > T$$

∴ The circuit is OK.



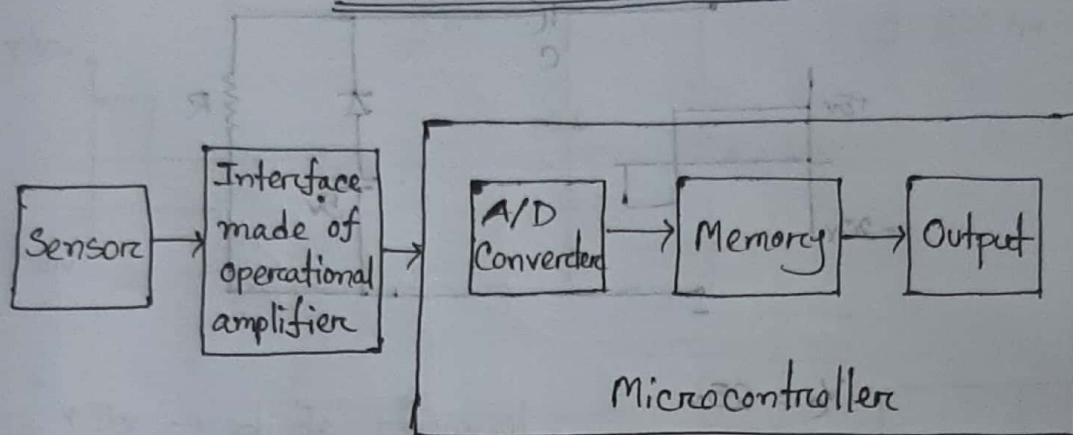
$$-5v - 25v + V_R = 0$$

Output: $\Rightarrow V_R = 30v$

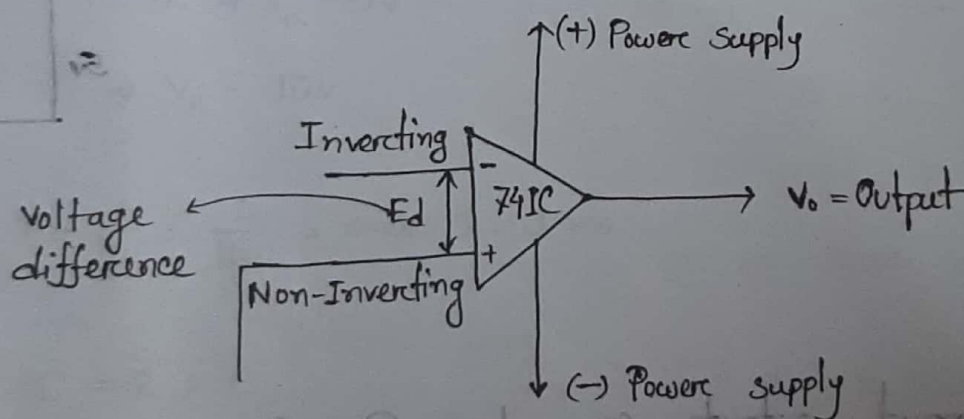


** अब input, output ko value diji dil R, C ko value
 (यह बयान है)

** Operation of clamper.

Operational Amplifier

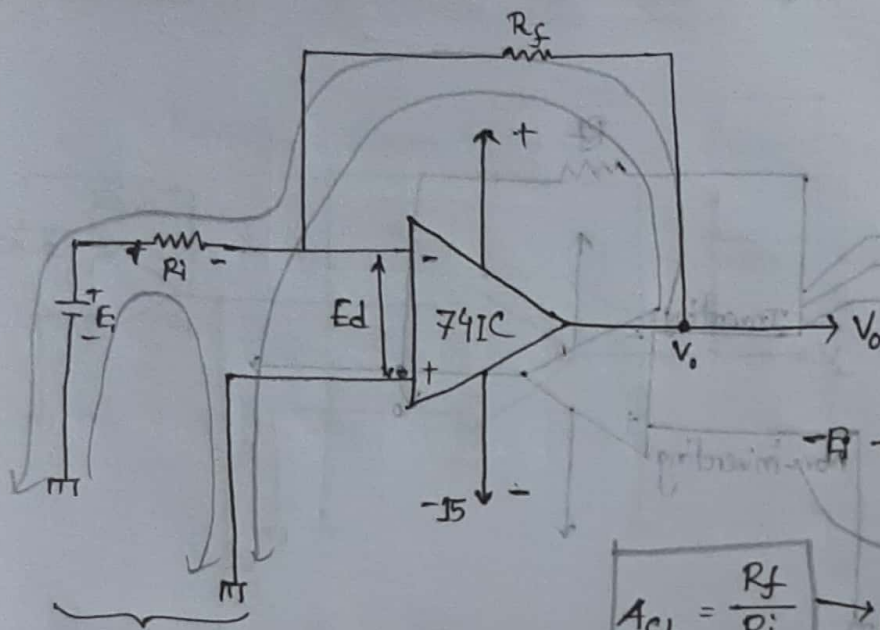
Op-amp: Interface between sensor & microcontroller



$E_d = \text{voltage at } +(\text{ve}) \text{ terminal} - \text{voltage at } -(\text{ve}) \text{ terminal}$

$$V_o = A_{OL} \times E_d$$

OL = Open Loop gain
CL = Close Loop gain



R_f = Feedback Resistance

R_i = Input Resistance

$$-E_i + I R_i = 0$$

$$I = \frac{E_i}{R_i}$$

$$A_{cl} = \frac{R_f}{R_i}$$

Efficient factor

2 terminal E_i
difference zero
shul closed loop
keb chul shul

$$\begin{aligned} V_{R_f} &= I R_f \\ &= \frac{E_i}{R_i} \times R_f \\ &= E_i \times \frac{R_f}{R_i} \end{aligned}$$

$$\begin{aligned} V_o &= -\frac{E_i}{R_i} \times R_f \\ &= -E_i \times \frac{R_f}{R_i} \end{aligned}$$

input

$V_o = -\frac{E_i}{R_i} \times R_f$
→ चेला दिव
आक याव जाव
 V_o, V_{R_f} का
negative

Work = Technical term of voltage

Parallel → same voltage
series → same current

** Example: Suppose $R_f = 100k\Omega$, $R_i = 10k\Omega$, $E_i = 2V$. Find out

i) I ; ii) A_{cl} ; iii) V_o

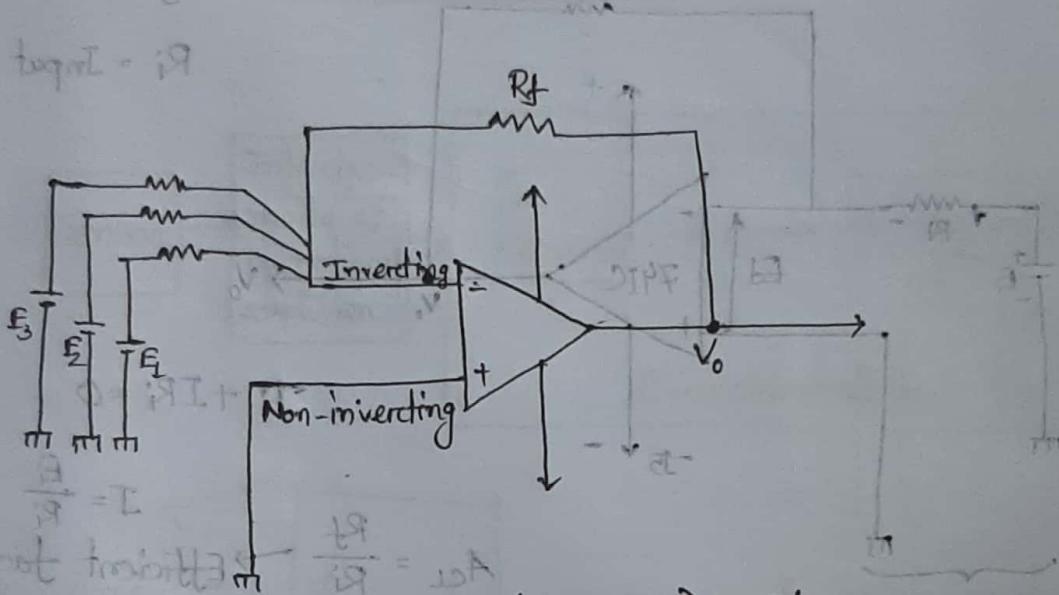
Solution:

$$\begin{aligned} \text{i) } I &= \frac{E_i}{R_i} \\ &= \frac{2}{10 \times 1000} \\ &= 0.0002 A \end{aligned}$$

$$\begin{aligned} \text{ii) } A_{cl} &= \frac{R_f}{R_i} \\ &= \frac{100k\Omega}{10k\Omega} \\ &= 10 \end{aligned}$$

$$\begin{aligned} \text{iii) } V_o &= -E_i \times \frac{R_f}{R_i} \\ &= -2V \times \frac{100 \times 1000}{10 \times 1000} \\ &= -20V \end{aligned}$$

Inverting Adder

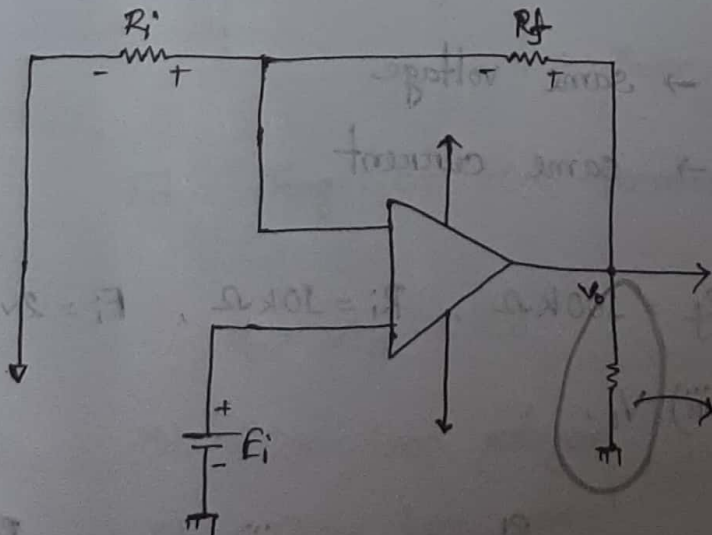


$$V_{Rf} = (I_1 + I_2 + I_3) \times R_f$$

$$V_i = -(I_1 + I_2 + I_3) \times R_f$$

$$= -\left(\frac{E_1}{R_1} + \frac{E_2}{R_2} + \frac{E_3}{R_3}\right) \times R_f$$

Non-Inverting Adder



यदि V_0 (यहाँ output) निकल रहा है

$$V_0 = V_{Rf} + E_i$$

$$= I R_f + E_i$$

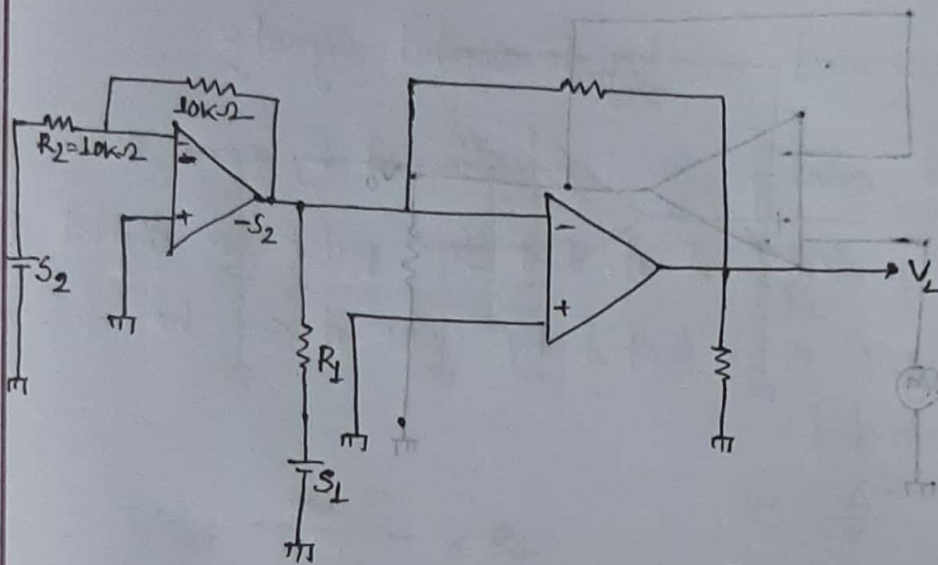
$$= \frac{E_i}{R_1} R_f + E_i$$

$$= E_i \left(1 + \frac{R_f}{R_1}\right)$$

$$E_i = I R_1$$

$$I = \frac{E_i}{R_1}$$

Subtractor



$S_1, S_2 = \text{input}$

$$\begin{aligned}
 V_1 &= -S_2 \cdot \frac{R_f}{R_i} \\
 &= -S_2 \cdot \frac{10k\Omega}{10k\Omega} \\
 &= -S_2
 \end{aligned}$$

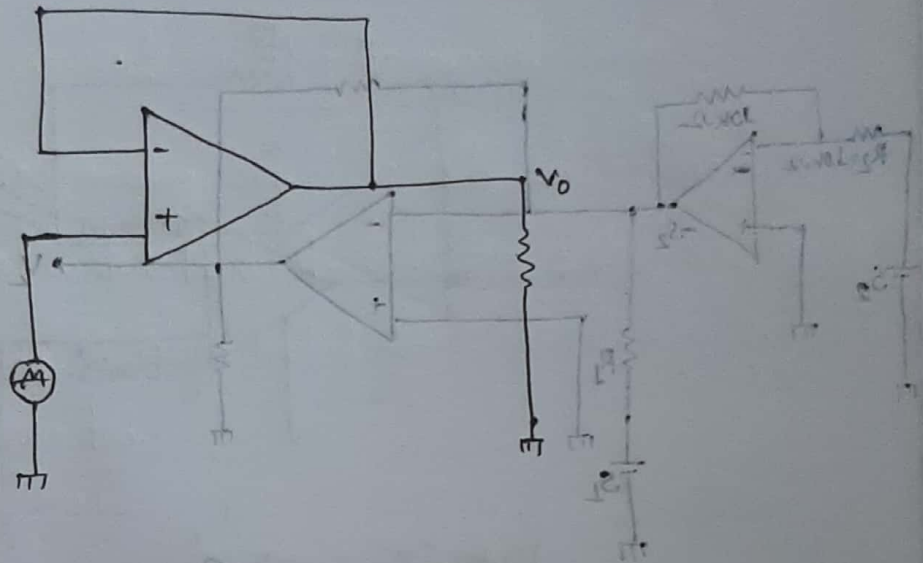
$$-S_2 + S_1$$

Non-inverting, $A_{CL} = 1$

$$\begin{aligned}
 A_{CL} &= \frac{V_o}{E_i} = 1 + \frac{R_f}{R_i} \\
 &= 1 + \frac{0}{R_i} \\
 &= 1 + 0 \\
 &= 1
 \end{aligned}$$

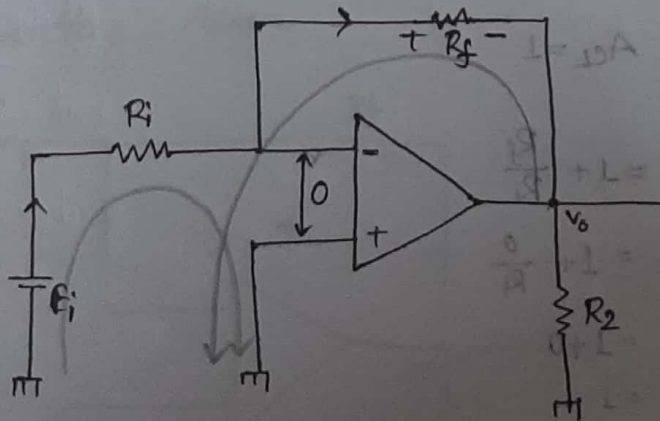
$$V_o = E_i \left(1 + \frac{R_f}{R_i} \right)$$

Voltage Follower / Unity Gain Amplifier



** क्या R_f को short कर सकते हैं?

$$A_{CL} = 1$$



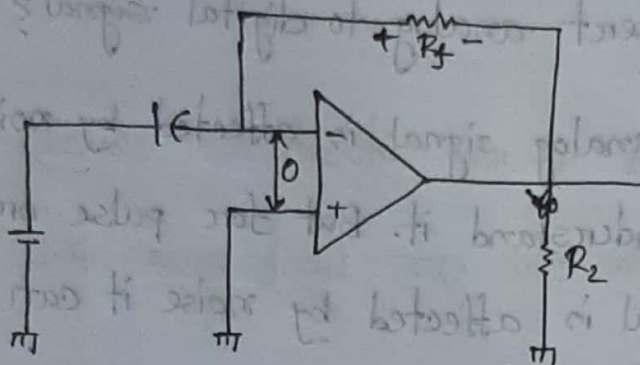
$$I = \frac{E_i}{R_i}$$

$$V_{R_f} = I R_f$$

$$= -\frac{E_i}{R_i} \times R_f$$

** What is V_{R_f} and how to determine it?

Differentiator



$$V_{Rf} = IR_{f}$$

$$V_0 = -C \times \frac{dE_i}{dt} \times R_f$$

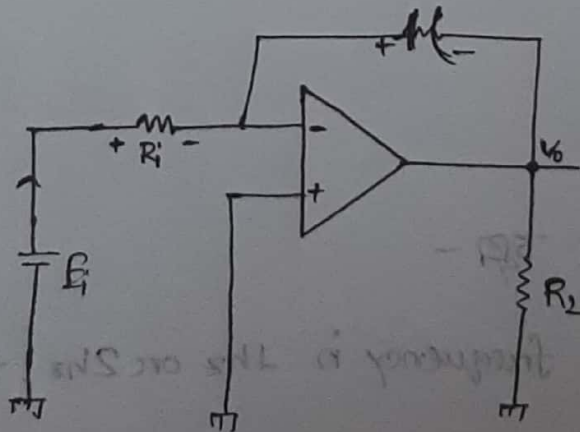
$$V_{Rf} = \frac{CdE_i}{dt} \times R_f$$

$$C = \frac{Q}{V} \rightarrow \text{charge}$$

$$\Rightarrow Q = C \times V$$

$$\frac{dQ}{dt} = C \times \frac{dE_i}{dt} = I$$

Integrator



$$I = \frac{E_i}{R_i}$$

$$V = \frac{Q}{C} = \frac{1}{C} \int \frac{E_i}{R_i} dt$$

$$= \frac{1}{R_i C} \int E_i dt$$

Current = charge at unit time

DAC & ADC-DAC

* Why we convert analog to digital signal?

➤ When the analog signal is affected by noise, then there's no way to understand it. But for pulse or digital signal, when the signal is affected by noise it can be detected as well as corrected.

*

* Flash ADC

* Waveform Digitizing.

- Sampling
- Quantize
- Encoding

Sample लेना व ड्यूटी -

"If signal frequency is 1hz or 2hz, then sampling must be 4hz minimum."

यदि 4hz को बड़ा है, यानी signal analog से digital बनाने पर receive करने से Digital to analog करे. जहाँ expected signal प्राप्त हो जाय वा.
(Aliasing Effect)

Quantization error:

* How to reduce quantization error by fixing the quantizing level

⇒ 0.5, 1.5, 2.5 এভাবে fixed করে দিলে কিছু error কমে.

* DAC Circuits

[Pspice এ Op-amp এর চর করে problem solve করে দেখাতে হবে]

Digital Logic Families

Logic family: A group of compatible ICs with some logic level and supply voltages fabricated for performing various logic functions referred to as a Logic family.

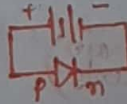
Semiconductor devices → 1. Bipolar: both type charges e^- & holes
2. Unipolar: one type charge carriers

Switching Property:

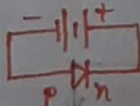
Diode

(Anode) P —  — N (Cathode)

F.B → Switch on

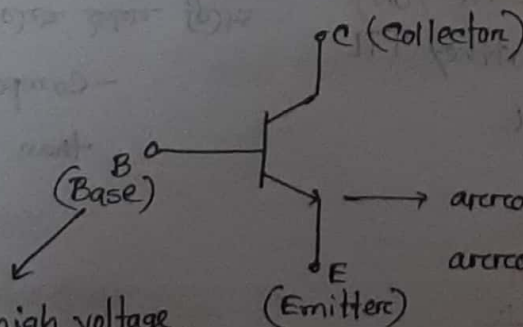


R.B → Switch off



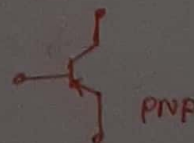
Switching property कहते हैं -
unidirectional property - यथा forward bias करा है तब switch on हो जायेगा वरना Reverse bias करा है तब switch off हो जायेगा।

Transistor



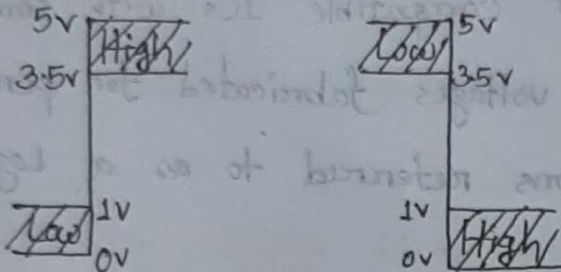
Base on high voltage
apply करके → ON switch
Low voltage → OFF switch

arrow वाले कि रहल NPN transistor
arrow वाले कि रहल PNP transistor



Compatible ICs $\rightarrow$ similar type ICs

Logic level: આ logic family-ya હજી જાણનારા



Low i/p $\rightarrow$ High o/p
High i/p $\rightarrow$ Low o/p

Positive Logic

Negative Logic

High i/p $\rightarrow$ High o/p
Low i/p $\rightarrow$ Low o/p

Classification:

Digital Logic Families



Bipolar L.F./IC (registers, diodes, transistors)

Unipolar L.F (MOS Devices)

BiCMOS L.F. (Most Bipolar Devices)

1. PMOS (P channel MOSFET)
2. NMOS (N channel MOSFET)
3. CMOS (P & N)

$\rightarrow$ i/p & logic operation = CMOS
 $\rightarrow$ o/p = Bipolar

(operation)
Saturated (saturation $\rightarrow$ ON)
Cut off $\rightarrow$ OFF

(operation)
Non-Saturated (Active $\rightarrow$ ON)

(Active region & cut off region ko નાલી વાજ વગેરે transistor)

- comparatively faster than saturated family

1. RTL
2. DTL
3. DCTL
4. I²L
5. HTL
6. TTL

1. Schottky TTL
2. ECL

Transistors are driven to saturation

transistor as a switch
વગર વગર &
cut off and saturated region ko નાલી વાજ વગર

Characteristics of Digital ICs :

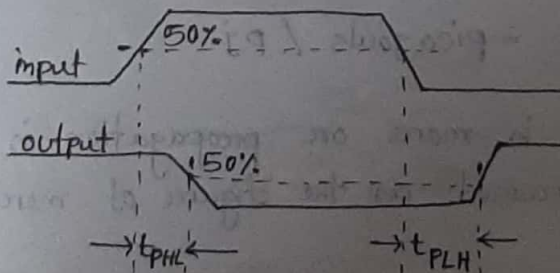
1. Propagation Delay
2. Power Dissipation
3. Figure of Merit
4. Fan-Out
5. Noise-immunity
6. Operating Temperature
7. Current & voltage parameters

Propagation Delay: of a gate is the average transition time that its signal takes to propagate from input to the output.

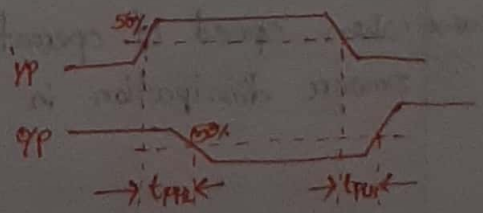
{ in nano sec.
 10^{-9} s }

- speed of operation is specified in terms of propagation delay. (Low prop. delay = high speed)

- Delay time are measured between 50% of input and output transitions



$$\text{Average prop. delay} = \frac{t_{PHL} + t_{PLH}}{2}$$



t_{PHL} = delay time when o/p goes high ; t_{PLH} = delay time when o/p goes low to high

✓ Power Dissipation: is the amount of power dissipated in an IC in milli watts in the form of heat.
mW

$$P_D = V_{CC} \times I_{CC}$$

- excessive heat/temp. can damage the IC, So, power dissipation should be less
- determined by the current ~~IE~~ I_{CC} that is drawn from V_{CC}

$$I_{CC} (Avg) = \frac{I_{CCH} + I_{CCL}}{2}$$

$$P_D = V_{CC} \times I_{CC}$$

V_{CC} for TTL IC
 V_{DD} for CMOS IC

I_{CCH} = current drawn from the power supply when o/p voltage is at high level

$$P_D (Avg) = V_{CC} \times I_{CC} (Avg) \\ = V_{CC} \times \frac{I_{CCH} + I_{CCL}}{2}$$

I_{CCL} = at low level

✓ Figure of Merit: is defined as the product of speed and power.
(in picojoules)

- should be as low as possible

$$\text{Figure of Merit} = \text{propagation delay} \times \text{Power}$$

$$= ns \times mW$$

$$= 10^{-9} s \times 10^{-3} W$$

$$= 10^{-12} \text{ Joules}$$

$$= \text{picojoules} / PJ$$

*** when speed of operation is more or propagation is less then the power dissipation is increased as the figure of merit remains constant

Fan-Out: no. of similar gates which can be driven by a gate.

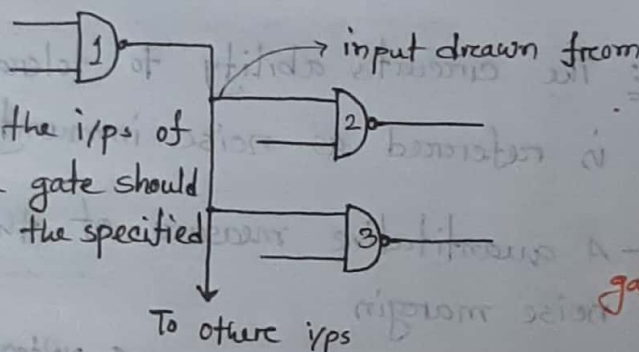
- higher the fanout, higher the current supplying capacity of a gate

- large no. of fanout means, more no. of gates can be driven from a gate

- slower for more fanouts

Fan-In: The no. of i/p's of a gate that it can handle without impairing its normal operation.

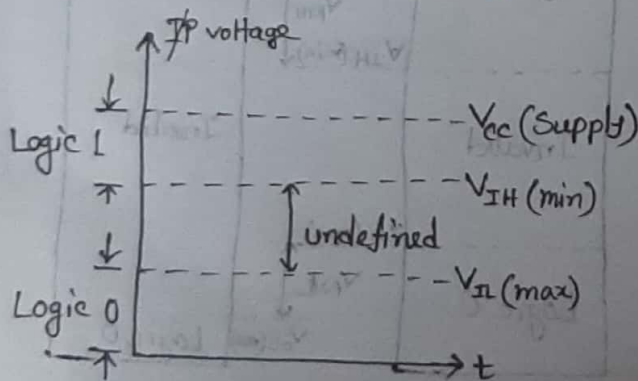
while driving the i/p's of these gates, the gate should not fall outside the specified voltage limits



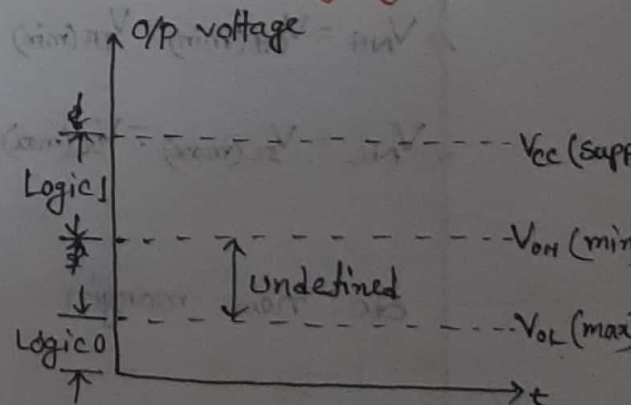
gate with larger fan-in are bigger and slower.

Voltage parameter:

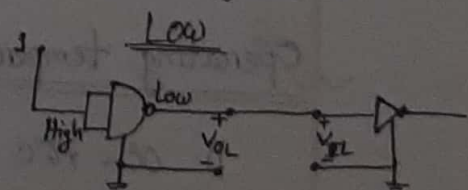
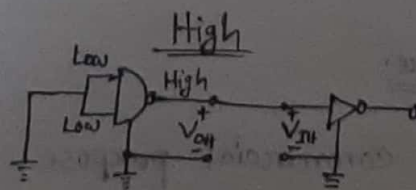
Voltages are specified for IC designing.



Input voltage parameters



Output voltage parameters



$V_{IL}(\max)$ - low level i/p voltage

$V_{OL}(\max)$ - low level o/p voltage

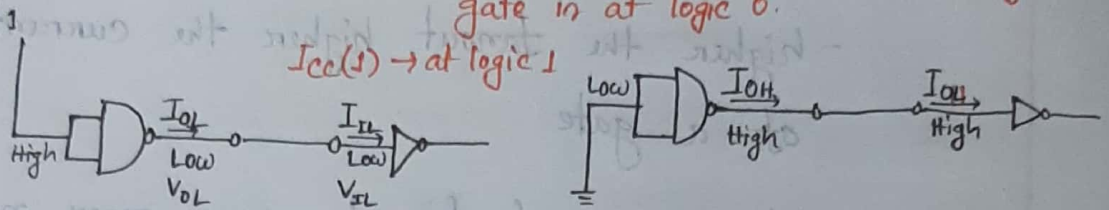
$V_{IH}(\min)$ - high level i/p voltage

$V_{OH}(\min)$ - high level o/p voltage

Current Parameters:

$I_{CC}(0) \rightarrow$ supply current when o/p of logic gate is at logic 0.

$I_{CC}(1) \rightarrow$ at logic 1

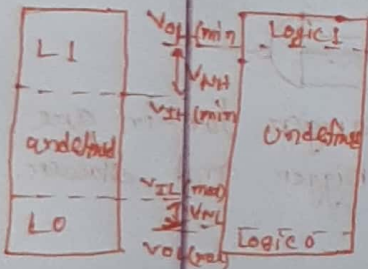


I_{IL} - low level i/p current ; I_{IH} - high level i/p current

I_{OL} - low level o/p current ; I_{OH} - high level o/p current

Noise immunity: The circuit's ability to tolerate noise signals is referred as noise immunity.

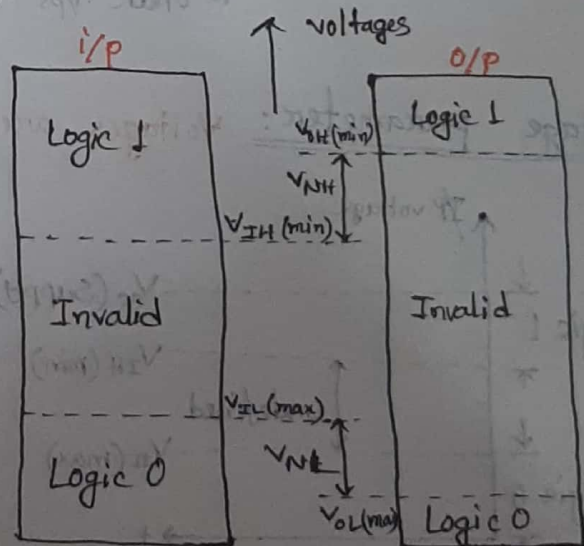
- A quantitative measure of this is called noise margin



$$V_{NH} = V_{OH}(\min) - V_{IH}(\min)$$

$$V_{NL} = V_{IL}(\max) - V_{OL}(\max)$$

dc. noise margin



voltage levels and noise margins of ICs

Operating temperature:

i) $0^\circ\text{C} - 70^\circ\text{C} \rightarrow$ commercial purpose

ii) $-55^\circ\text{C} - 125^\circ\text{C} \rightarrow$ military application

Resistor Transistor Logic (RTL): is a class of digital circuits built using resistors as the input network and bipolar junction circuit transistors as switching devices.

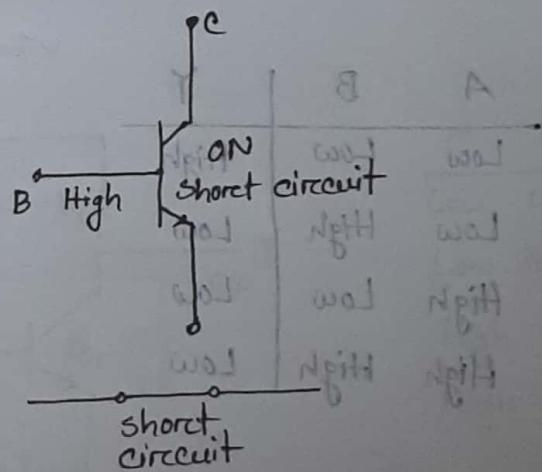
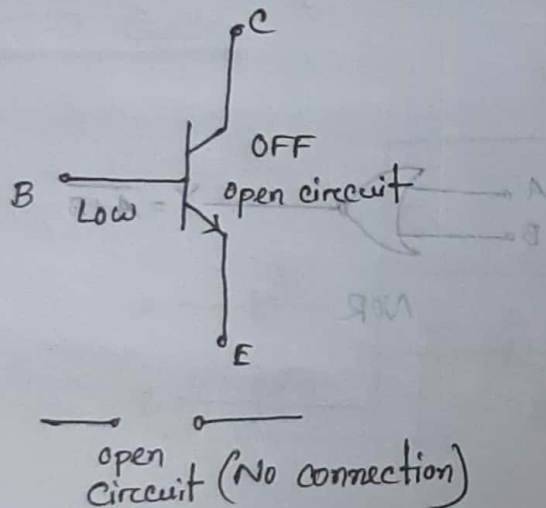
Switching mechanism: (of transistor)

1. Cut-off region $\rightarrow$ OFF state
2. Saturation region $\rightarrow$ ON state

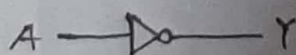
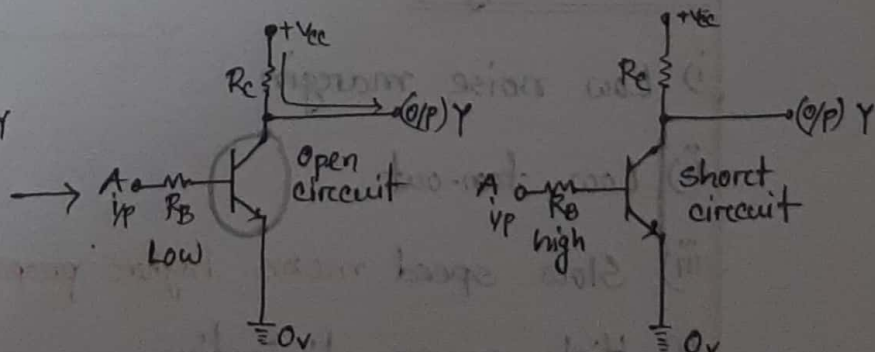
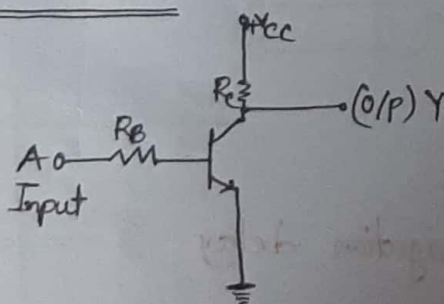
RTL specifications:

Low: 0.2 V

High: 1 to 3.6 V

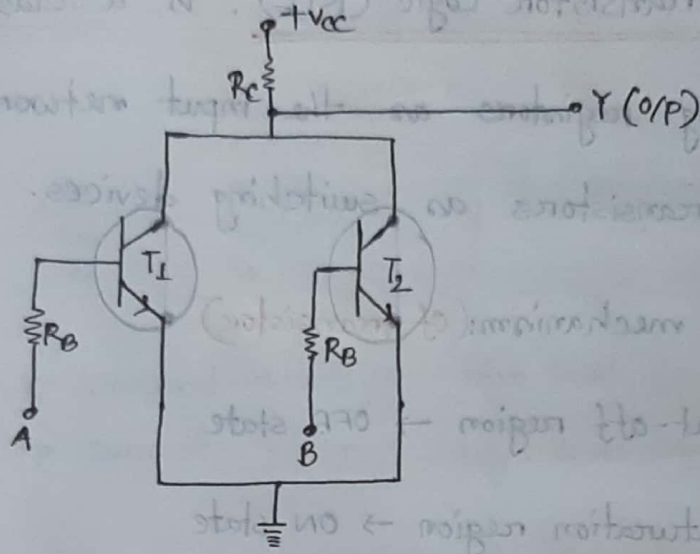


NOT Gate:

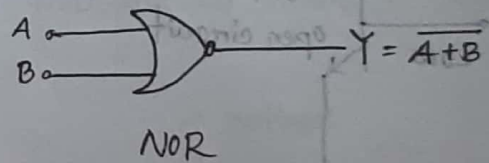


I/P (A)	O/P (Y)
Low	High
High	Low

NOR Gate:



A	B	Y
Low	Low	High
Low	High	Low
High	Low	Low
High	High	Low



Disadvantages:

- i) Low noise margin
- ii) Low fan-out
- iii) Slow speed means higher propagation delay
- iv) High power dissipation

Diode Transistor Logic (DTL): refers to the technology for designing and fabricating digital circuits wherein logic gates employ diodes in the input stage and bipolar junction transistors at the output stage.

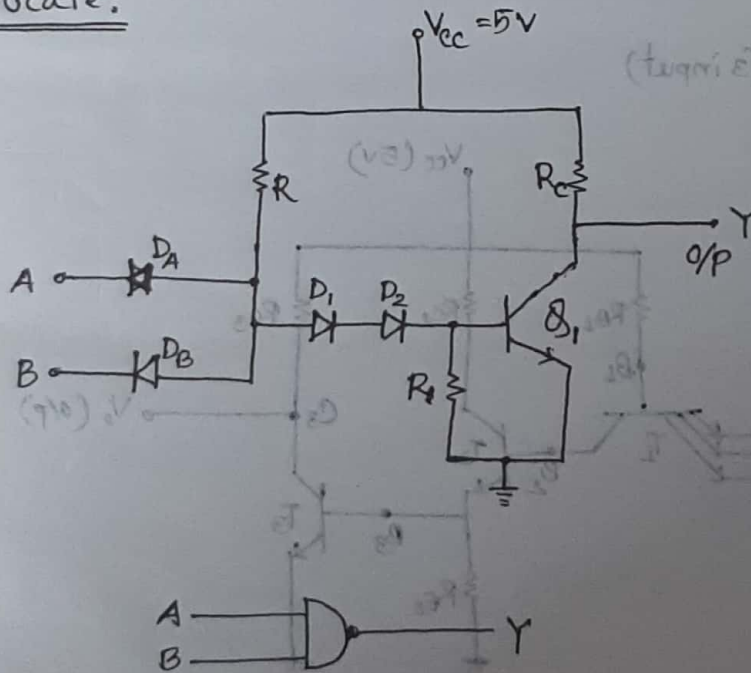
- Greater fan-out
- Improved Noise-margin

DTL Specifications

Low: 0.2V

High: 4-5V

NAND Gate:

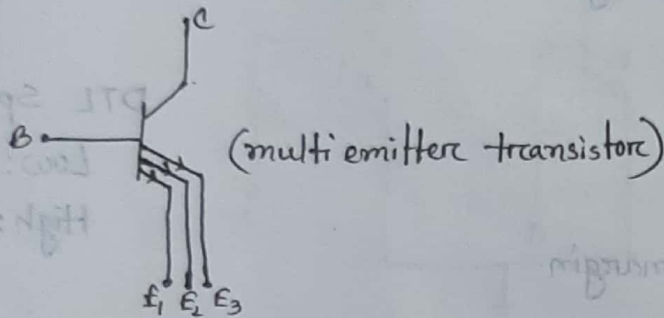


A	B	Y
Low	Low	High
Low	High	High
High	Low	High
High	High	Low

Disadvantage:

- Slow speed
- More propagation delay

Transistor-Transistor Logic: is a logic family built from bipolar junction transistors where transistors perform both the logic function and amplifying function.



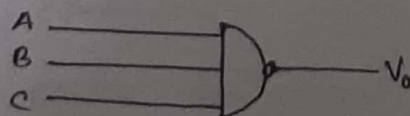
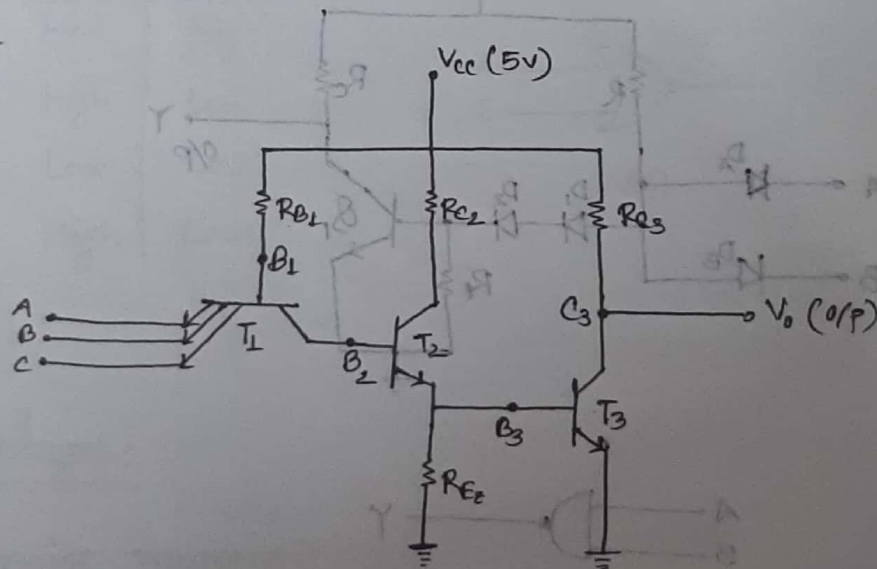
TTL Specifications

Logic 0/Low = $0.2V (V_{CEsat})$

Logic 1/High = $5V$

propagation delay = $\frac{1}{10} t_{TL}$

NAND Gate: (3 input)



A	B	C	V ₀
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Disadvantages:

- i) Consumes more power compared to CMOS
- ii) It has 10 fanout which is less than CMOS

RTL, DTL, TTL

* 3-Input RTL NOR gate with working principle.

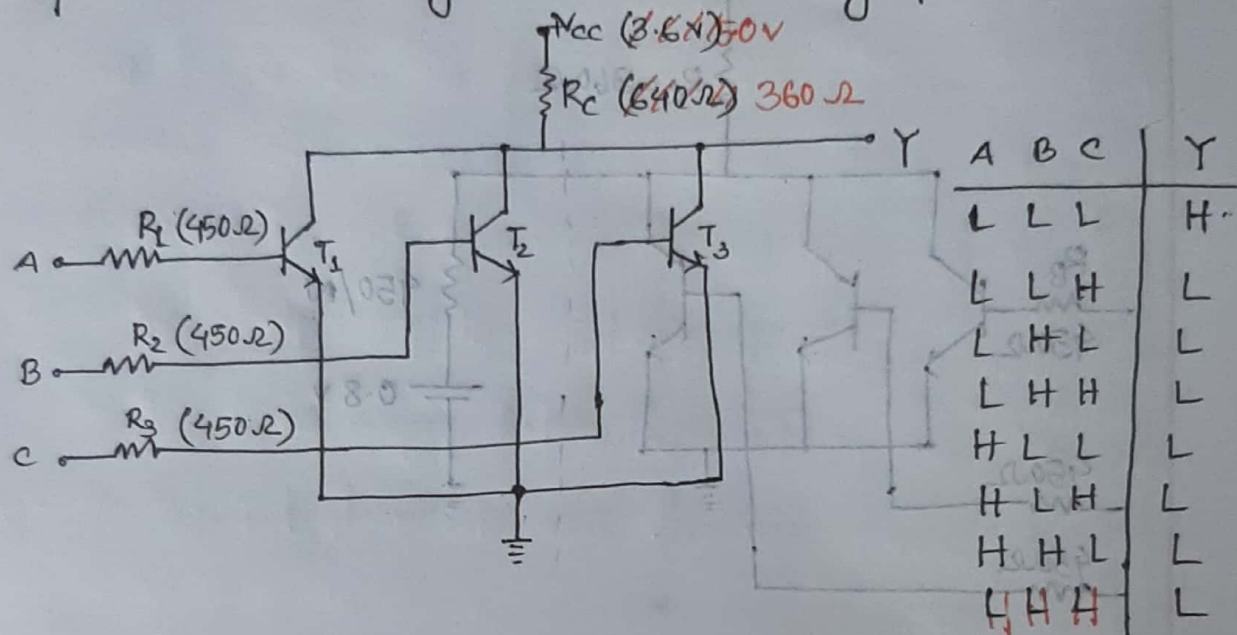


Figure: 3-input RTL NOR gate

The figure above shows that T_1, T_2, T_3 are three transistors and A, B, C are the three inputs, given to the base of the three transistors. Y is the output.

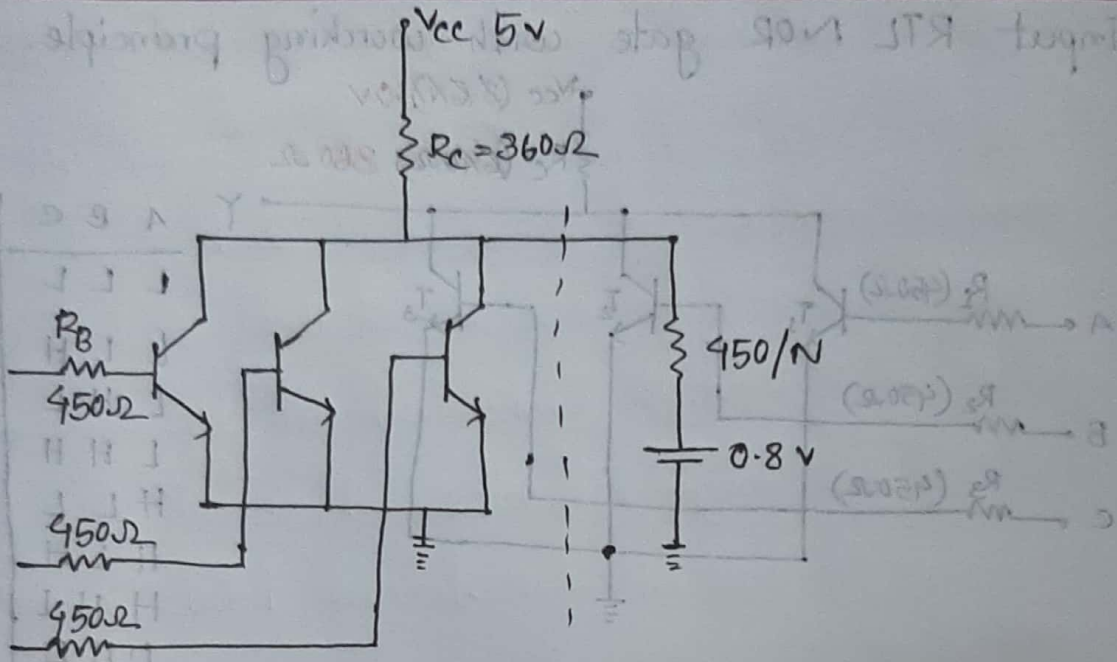
When all the three inputs are low or 0, the transistors are in the cut off region. The collector and emitter terminal will act as an open circuit and the V_{cc} will pass through the output, making the output high or 1.

When any of the ~~trans~~ input is high or 1, the transistor will be in its saturated region and will act as the e and c terminal will act as short circuit. So, the current will flow it will make a path for the supply voltage V_{cc} to go to the ground through the resistor R_c and transistor. Thus there will be 0 or low at o/p terminal Y .

$h_{FE} = 10, N = 5, V_{CC} = 5V, \text{cut-in voltage} = 0.5V$

$R_B = 450\Omega, R_C = 360\Omega$

**



Now,

$$I_B = \frac{5 - 0.8}{360 + \frac{450}{5}} \cdot \frac{1}{5} = 0.00186 A = 1.86 mA$$

$$I_C = \frac{5 - 0.2}{360} = 0.0133 A = 13.3 mA$$

The value of N must satisfy the relation

$$h_{FE} \cdot I_B \geq I_C$$

$$\Rightarrow 10 \times 1.86 \geq 13.3$$

$$\Rightarrow 18.6 \geq 13.3$$

$N=5$ satisfies the equation.

Noise margin for logic 0,

$$\Delta 0 = 0.5 - 0.2 = 0.3$$

Noise margin for logic 1,

$$V_0 = \left(5 \times \frac{450/5}{360 + 450/5} \right) + \left(0.8 \times \frac{360}{360 + 450/5} \right) = 1.64$$

For $h_{FE} = 10$, V_0 must be $\left(5 \times \frac{13.3}{10} \right) mA = 6.65 mA = 0.00665 A$

Now, $(0.00665 \times 90 + 0.8) = 1.4$; $\therefore \Delta 1 = 1.64 - 1.4 = 0.24$

* 3-input DTL NAND Gate and its working principle.

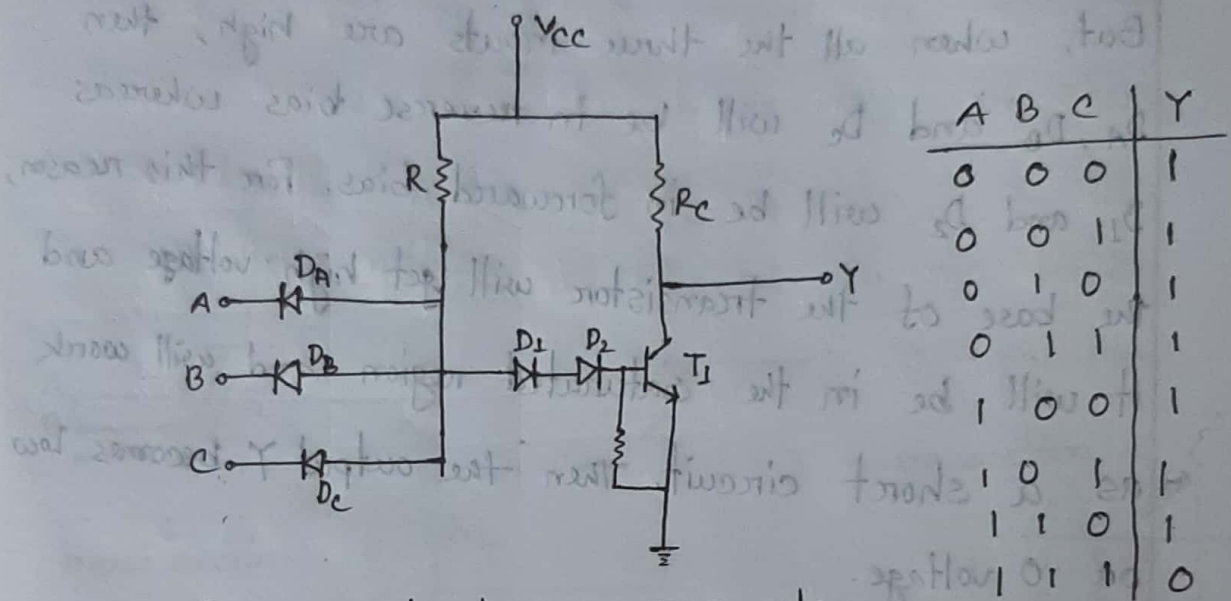


Figure- 3-input DTL NAND gate

When the inputs are low signal or 0, the D_A , D_B , D_C diodes will act as forward bias, whereas D_1 and D_2 will act as reverse bias. As D_1 and D_2 are working as reverse bias, the base current of transistor T_1 will be low, means the transistor will be in the cut off region and will act as an open circuit. So the output will be V_{cc} which will pass through the R_c .

Again, when one of the input will be high or 1, then one of the three diodes will be forward biased and others will be in reverse biased. So then D_1 and D_2 will be in the reverse bias. As a result the base of the transistor will get low voltage or 0. Then the transistor will be in the cut off region.

Hence V_{cc} or high voltage will be the output.

But, when all the three inputs are high, then D_A , D_B and D_C will be in reverse bias whereas D_1 and D_2 will be in forward bias. For this reason, the base of the transistor will get high voltage and it will be in the saturated region and will work as a short circuit. Then the output Y becomes low or 0 voltage.

Figure 3-input DTL NAND Gate

* 3-input NAND TTL NAND and its working principle:

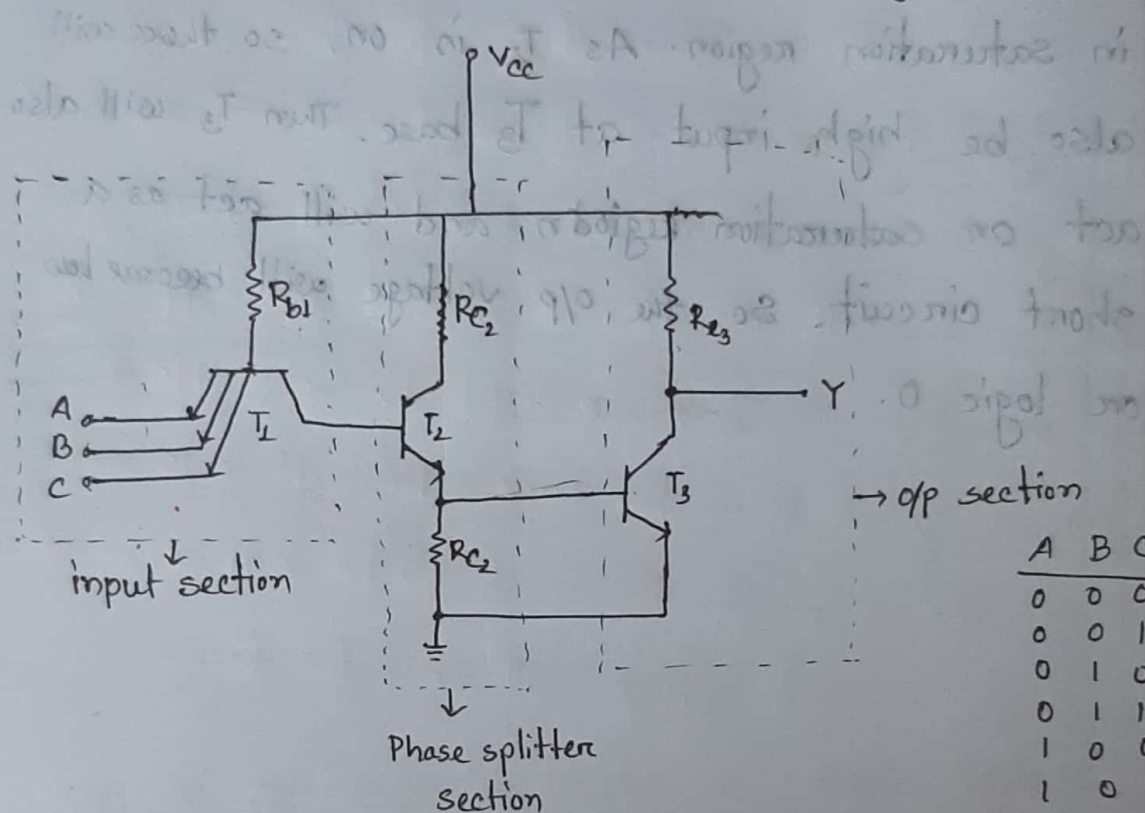


Figure: 3-input TTL NAND

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

When any of the input is low means logic 0, the base-emitter for T_1 transistor will act as forward bias and base-collector will work as reverse bias. For this reason, base input for T_2 will be low, so it will be in cut off region. As T_2 is in cut off region, T_3 will also be in cut off region. So, the output voltage will be high or 1.

On the other hand, when the inputs are all 1 or high, the base-emitter for T_1 will work as reverse bias and base collector will work as forward bias. For this reason, base

input for T_2 will become high and T_2 will act in saturation region. As T_2 is on, so there will also be high input at T_3 base. Then T_3 will also act on saturation region and will act as a short circuit. So, the o/p voltage will become low or logic 0.

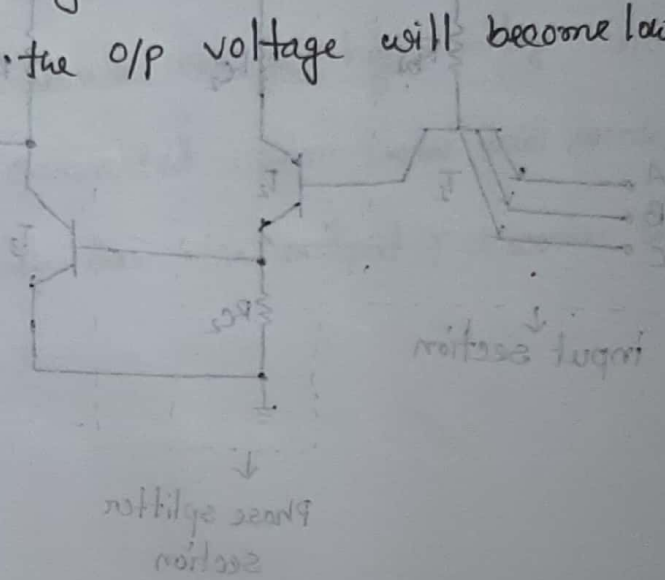


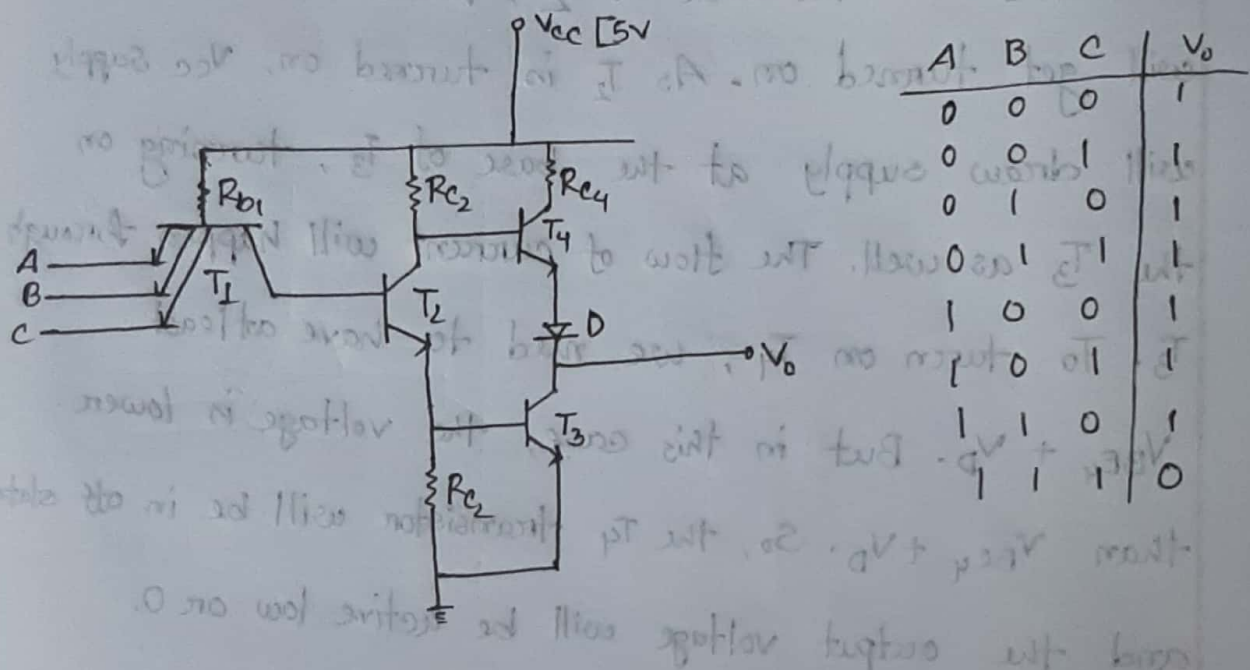
Figure: 3-input TTL NAND

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

When any of the input is low means logic 0, the base of T_1 transistor will act as forward bias and the emitter of T_1 transistor will be in reverse bias. For this reason, base input of T_1 will be low, so it will be in cut off region. As T_1 is in cut off region, it will also be in cut off region. So, the output voltage will be high or 1.

On the other hand, when the inputs are all 1 or high, the base-emitter junction of T_1 will be in forward bias and the collector will be in reverse bias. For this reason, the collector will be in forward bias and the emitter will be in reverse bias. So, the output voltage will be low or 0.

* TTL NAND Gate with Totem pole Output



If any of the input is active low, means logic 0, In that case, the base-emitter junction of T_1 will be forward biased and base-collector junction will be in reverse bias. T_1 will be in the active region and open circuited at base collector junction. So, no input will be passed on base of T_2 , so T_2 will be off and open circuited. So, this will not allow supply to be given at the base of T_3 . Eventually T_3 will be also in off state. Supply voltage will be passed on the base of T_4 and T_4 will be in on state and will be short circuited. So, to calculate this o/p voltage, $V_o = V_{CC} - I_C R_{c4}(\text{sat}) - V_D$.

When all the inputs are provided with logic 1, the base-emitter junction will be in reverse bias and the base collector junction

will be in forward bias. So, the V_{cc} will draw a supply at the base of T_2 , the transistor T_2 will get turned on. As T_2 is turned on, V_{cc} supply will draw supply at the base of T_3 , turning on the T_3 as well. The flow of current will happen through T_3 . To turn on T_4 , we need to have at least $V_{BE4} + V_D$. But in this case, the voltage is lower than $V_{BE4} + V_D$. So, the T_4 transistor will be in off state and the output voltage will be active low or 0.

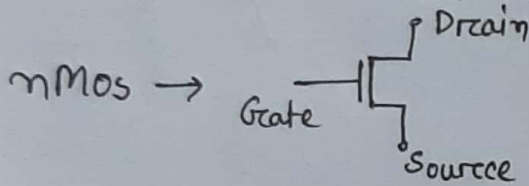
To turn on T_4 , when all input is 1, we need $0.7 + 0.7 = 1.4V$. But as T_2 and T_3 are in saturated mode, the voltage supply will become

$$V_{CE2} + V_{BE3} = 0.2 + 0.7 = 0.9V$$

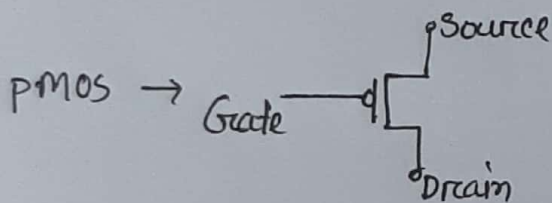
So, T_4 will turn off.

CMOS

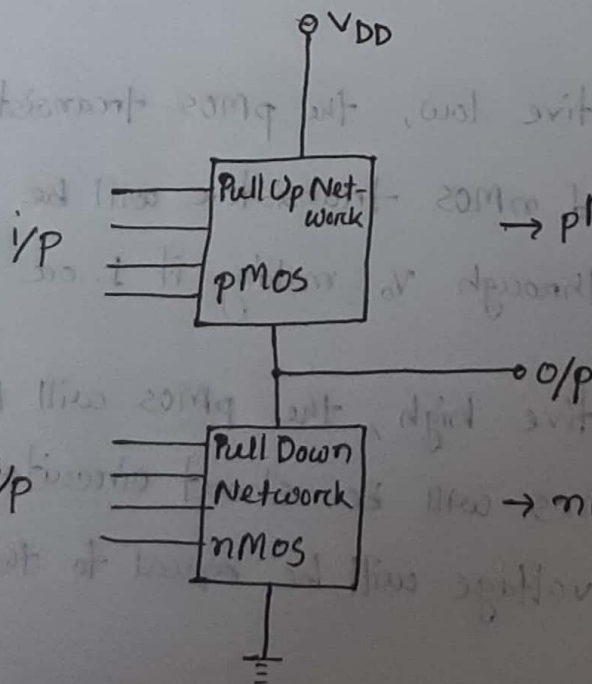
- Complement ~~n~~ Metal Oxide Semiconductor
- have 2 types of transistors i) PMOS ii) NMOS



- if $G_c = 0$, OFF (D to S as Open Circuit)
- if $G_c = 1$, ON (D to S as short circuit)



- if $G_c = 0$, ON (S to D as short circuit)
- if $G_c = 1$, OFF (S to D as open circuit)



→ pMOS transistor can easily pass Logic high

→ nMOS transistor can easily pass Logic low

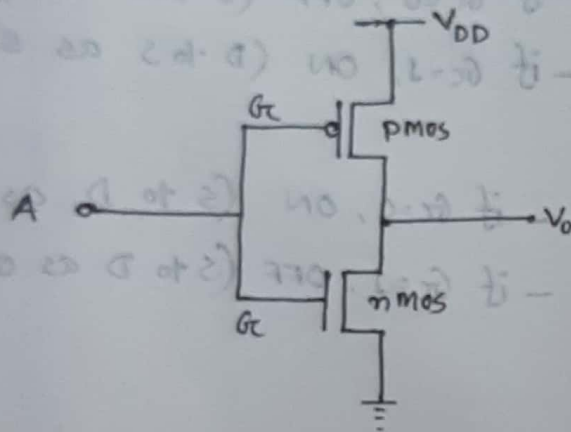
Fig: Block Diagram of CMOS inverter

2 operations :

i) AND ($\cdot$) $\rightarrow$ pmos parallel, nmos series

ii) OR ($+$) $\rightarrow$ pmos series, nmos parallel

* Working Principle of CMOS inverter



When input is active low, the pmos transistor will be short circuited and nmos transistor will be open circuited. So, V_{DD} will pass through V_0 making it 1. or active high.

When input is active high, the pmos will be open circuited and nmos will be short circuited. So the resulting output voltage will be equal to the ground which is logic 0.

Applications:

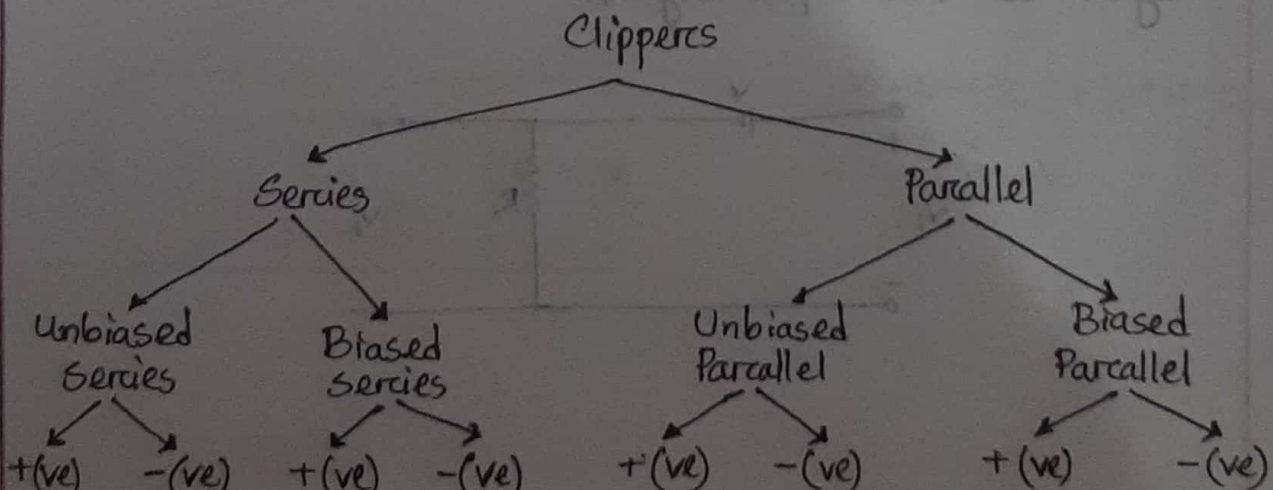
- used in different ICs like microprocessors, static RAM, microcontrollers, data converters, image sensors and transceivers.

Clippers & Clampers:

- We use diodes along with resistors and capacitors to shape waveforms.
- Clippers and clampers are called waveform shaping circuits.
- If we want to clip portion of the waveform, then we use clippers.
- If we want to shift or clamp the DC voltage level, then we use clampers.

Clippers

Clippers : are networks that use diodes to clip off a portion of the input signal without distorting the remaining part of the alternating waveform.



Series Configuration: The diode is in series with the load.

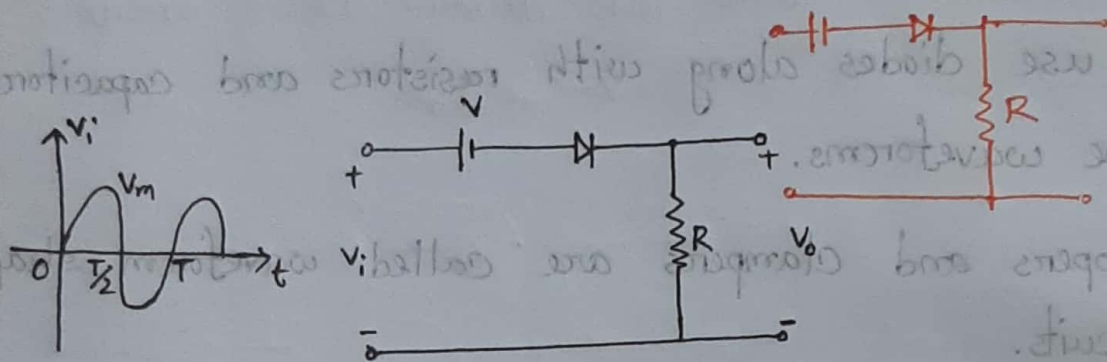
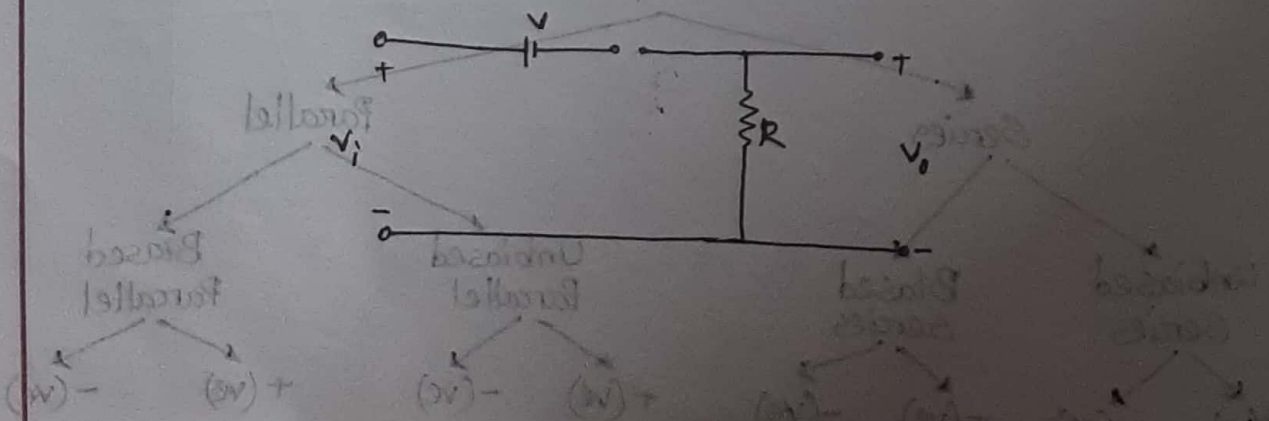
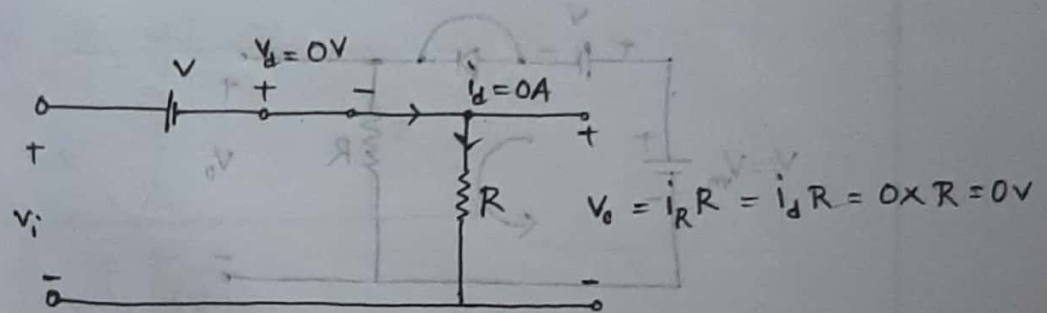


Fig: Series clipper with a dc supply

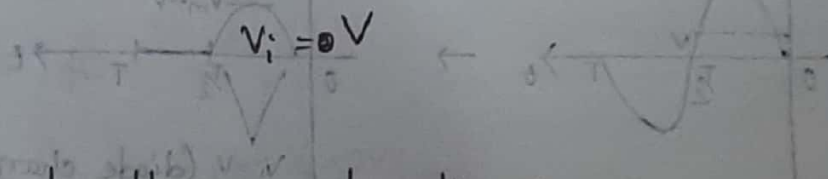
- The direction of the diode suggests that the signal v_i must be positive to turn it on.
- The dc supply requires that the voltage v_i be greater than V volts to turn the diode on.
- Negative region of the input signal pressurizing the diode into the off state, supported further by dc supply.
- The diode here, is an open circuit for the negative region of the input signal.



- For ideal diode, transition between states will occur at the point where $V_d = 0V$ and $i_d = 0A$. Applying this condition to the circuit, we got,

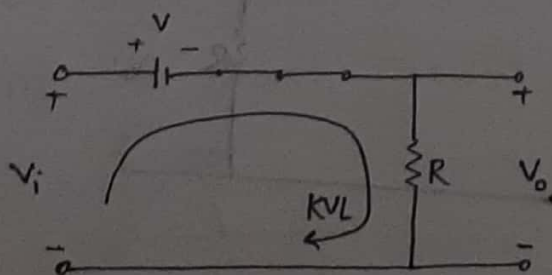


The transition level of v_i is,



For an input voltage greater than V volts, the diode is in short circuit state, and for input voltages less than V volts, it is in open circuit state.

- When the circuit is diode is in short circuit state, by applying Kirchhoff's voltage law, we can determine V_o .

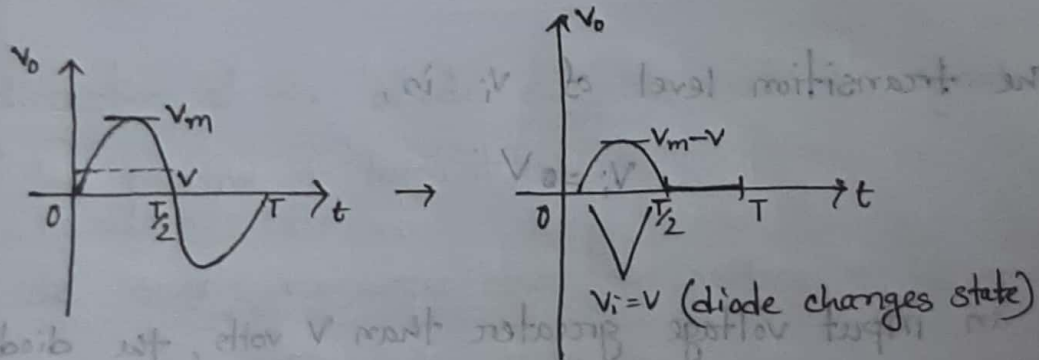
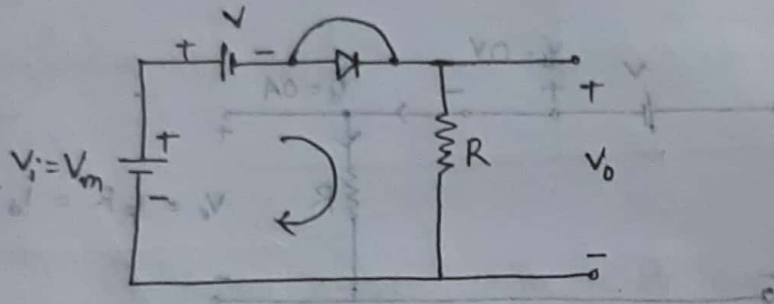


$$V_i - V - V_o = 0 \text{ (CW Direction)}$$

$$\Rightarrow V_o = V_i - V$$

For instance, at $V_i = V_m$ and for $V_m > V$ the diode is in short circuit state and

$$V_o = V_m - V$$



* Example 2.20: Determine the output waveform for the network of the given figure,

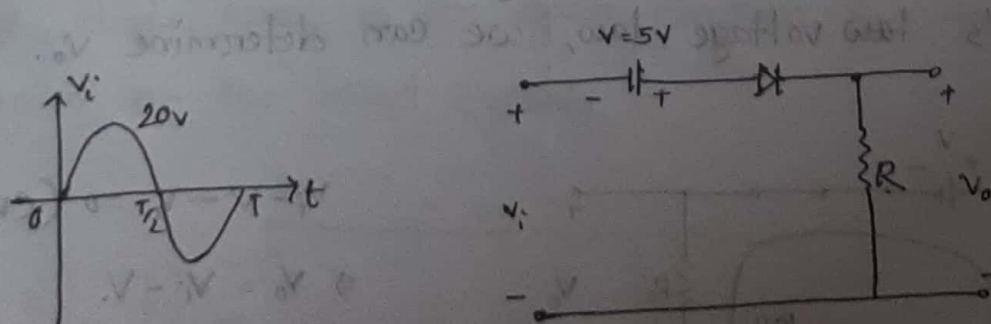
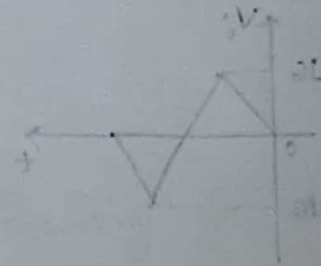
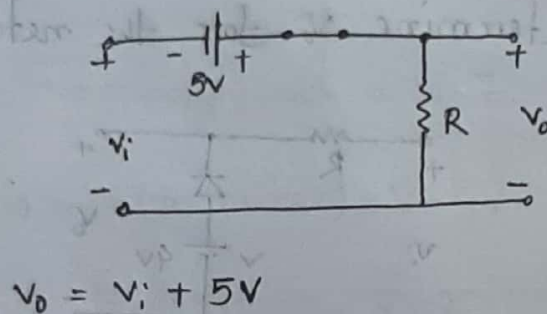


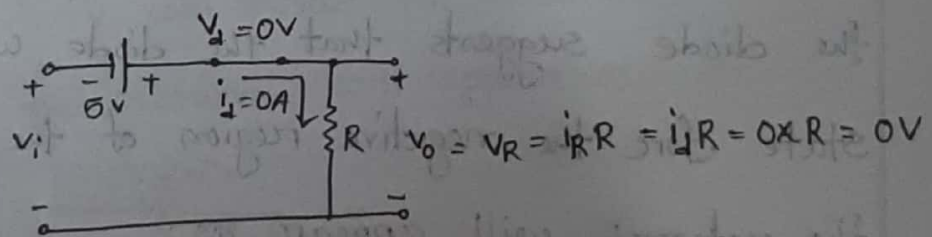
Figure- Series clipper

Solution: Parallel Configuration: The analysis of parallel circuit is as follows:

The diode will be in the 'on' state for the positive region of V_i especially when $V = 5V$. The network will then appear as:



Substituting $i_d = 0$ at $V_d = 0$ for the transition level, we obtain the network as:



$$\therefore V_i = -5V \quad (V_o = V_i + 5V \Rightarrow 0 = V_i + 5V \Rightarrow V_i = -5V)$$

For V_i more negative than $-5V$ the diode will enter its open-circuit state, while for voltage more positive than $-5V$ the diode is in the short circuit state.

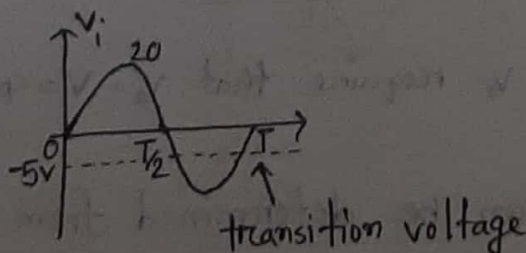


fig: input voltage

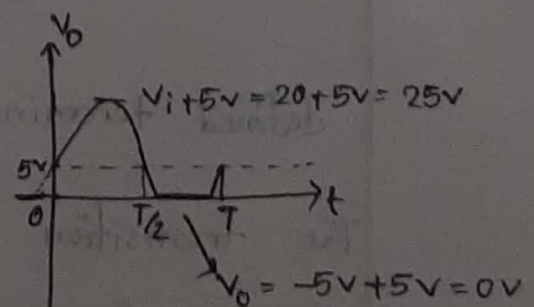
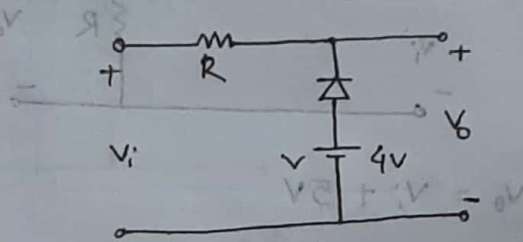
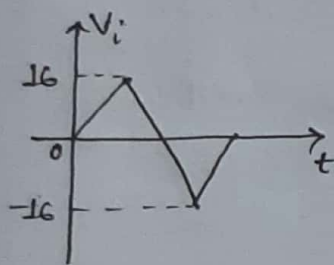


fig: output voltage

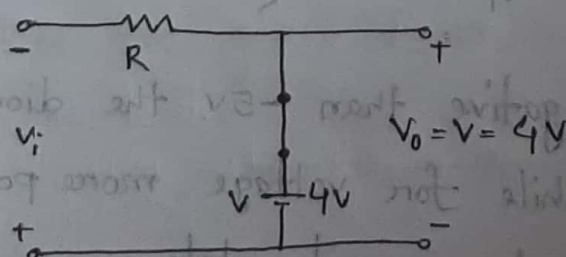
Parallel Configuration: The analysis of parallel configuration is very similar to that of applied to the series configuration.

* Example 2.22: Determine V_o for the network



Solution:

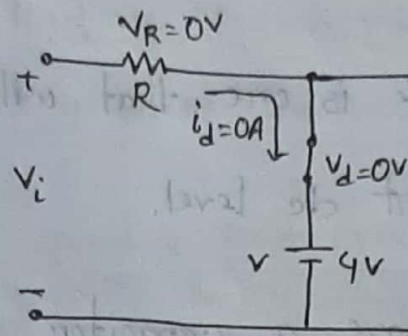
The polarity of the dc supply and the direction of the diode suggests that the diode will be in the on state for the negative region of the input signal. Then the network will appear as:



defined terminals for V_o require that $V_o = V = 4V$

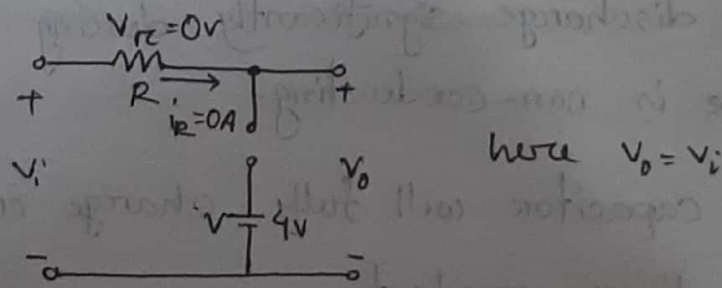
The transition state can be determined from the following figure where the condition $i_d = 0A$, $V_d = 0V$ has been

imposed.

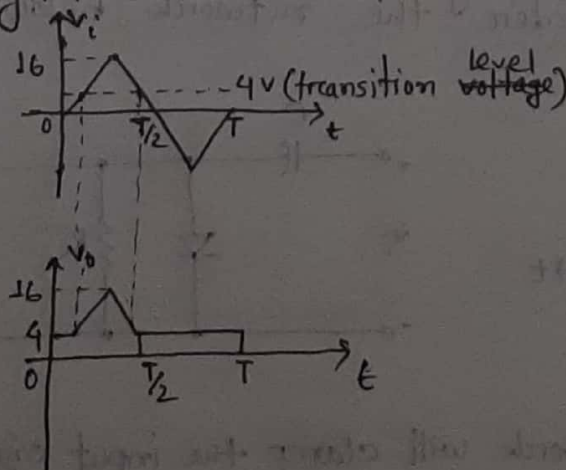


The result is $V_i(\text{transition}) = V = 4V$

Since the dc supply is obviously pressuring the diode to stay in the short-circuit state, input voltage must be ~~4~~ greater than 4V for the diode to be in the off state. Any input voltage ~~shorter~~ less than 4V will result in a short circuit. For open-circuit state, the network will appear as:



The output signal becomes as:

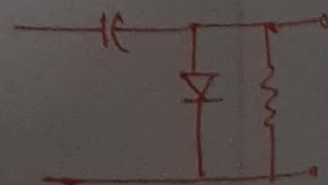
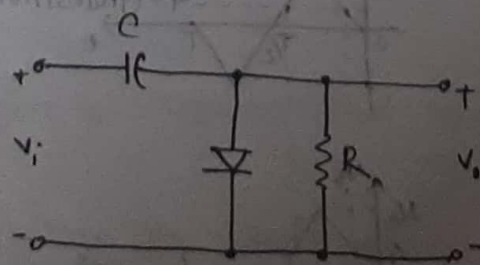
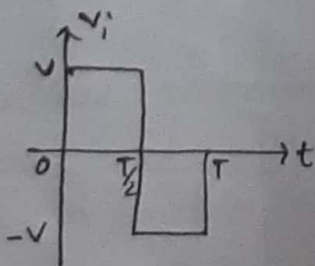


Clampercs

- The clamping network is one that will clamp a signal to a different dc level.
- The network must have a capacitor, a diode and a resistive element.
- It can also employ an independent dc supply to introduce an additional shift.
- The magnitude of R and C must be chosen such that the time constant $\tau = RC$ is large enough to ensure that the voltage across the capacitor does not discharge significantly during the interval the diode is non-conducting.
- The capacitor will fully charge or discharge in five times constant.

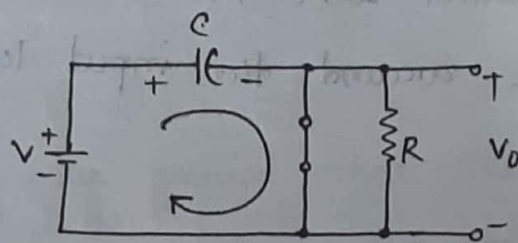
**** Design the clamping network and show the analysis of the network**

- Let us consider the network below:

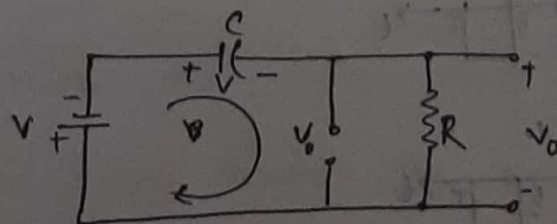


This input network will clamp the input signal to the zero level.

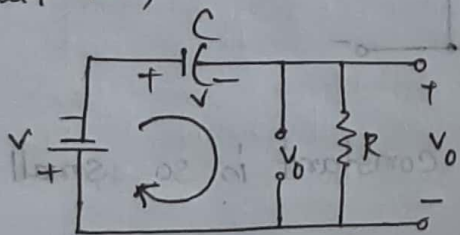
- The resistor R can be the load resistor or a parallel combination of the load resistor and a resistor designed to provide the desired level of R .
- During the positive half cycle ($0 \rightarrow T/2$) the network will appear as the following with the diode in the on state, shorting out the effect of the resistor R .



- The resulting RC time constant is so small that the capacitor will charge to V volts very quickly. During this interval the output voltage is directly across the short circuit and $V_o = 0V$.
- When the input switches to the $-V$ state, the network will appear as the following with the open circuit equivalent for diode.



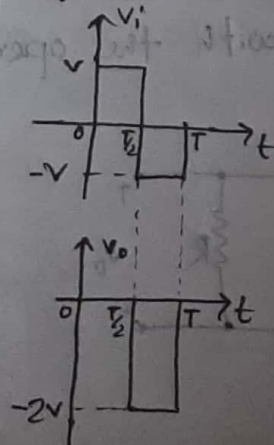
- Now that R is back in the network, the time constant determined by the RC product is sufficiently large to establish a discharge period 5τ , much greater than the period $T_2 \rightarrow T$.
- The capacitor holds onto all its charge and therefore voltage during this period.
- Since V_0 is parallel with the diode and resistor, applying the KVL around the input loop will result in,



$$-V - V - V_0 = 0$$

$$\Rightarrow V_0 = -2V$$

- The negative sign represents resulting from the fact that the polarity of $2V$ is opposite to the polarity defined for V_0 .

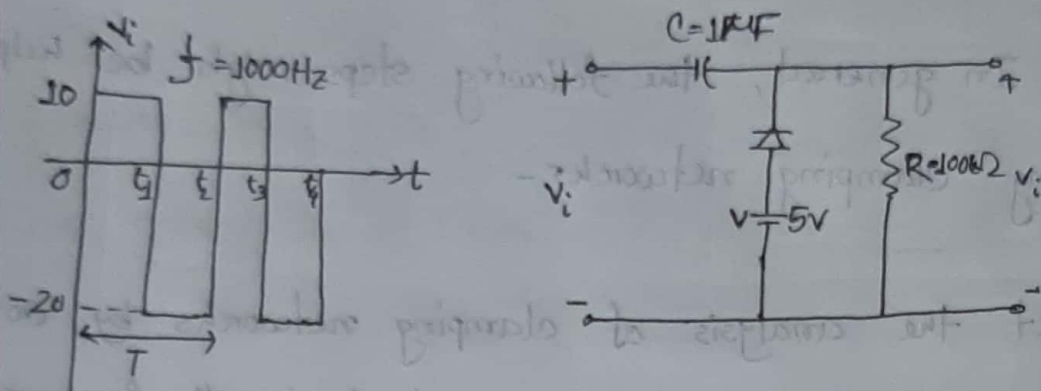


* Discuss about the steps that may be helpful when analyzing clamping networks.

Answer: In general, the following steps might be helpful when analyzing clamping networks -

1. Start the analysis of clamping networks by considering that part of the input signal that will forward bias the diode.
2. During the period that the diode is in on state, assume that the capacitor will charge up instantaneously to a voltage level determined by the network.
3. Assume that the period the diode is in the off state, the capacitor will hold onto its established voltage level.
4. Throughout the analysis maintain a continual awareness of the location and reference polarity for V_o to ensure that the proper levels for V_o are obtained.
5. Keep in mind the general rule that the total swing of the total output must match the swing of the input signal.

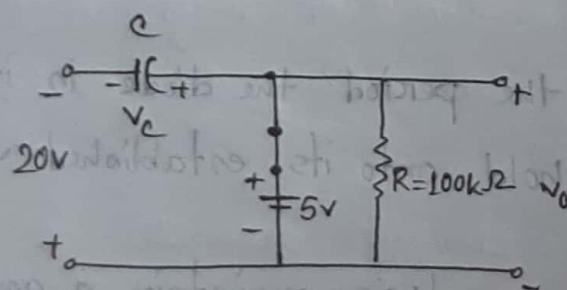
* Example 2.24: Determine V_o for the network of the following figure for the input indicated.



Solution:

Here, $f = 1000 \text{ Hz}$; So, $T = \frac{1}{f} = \frac{1}{1000} = 0.001 \text{ s} = 1 \text{ ms}$.

The analysis will begin with the period $t_1 \rightarrow t_2$ of the input signal since the diode is in its short circuit state.



The output is across R and it is also across the 5V battery. The result for this interval is $V_o = 5 \text{ V}$.

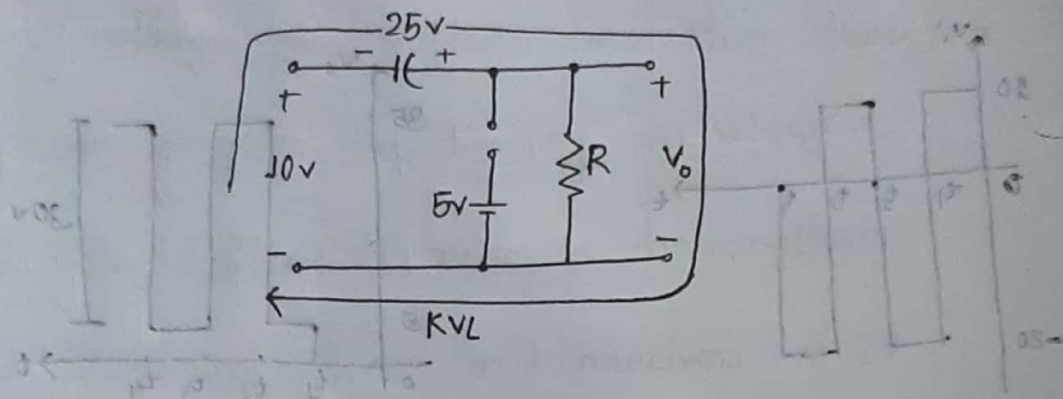
Applying KVL around the input loop will result in

$$-20 \text{ V} + V_c - 5 \text{ V} = 0$$

$$\Rightarrow V_c = 25 \text{ V}$$

The capacitor will charge up to 25 V.

For the period $t_2 \rightarrow t_3$ the network will appear as:



The open circuit equivalent for the diode will remove the 5V battery from having any effect on V_0 and applying KVL around the outside loop of network will result in

$$+10V + 25V - V_0 = 0$$

$$\Rightarrow V_0 = 35V$$

The time constant of the discharging network is determined by the product RC and has the magnitude of

$$\tau = RC = (100K\Omega)(0.1\mu F)$$

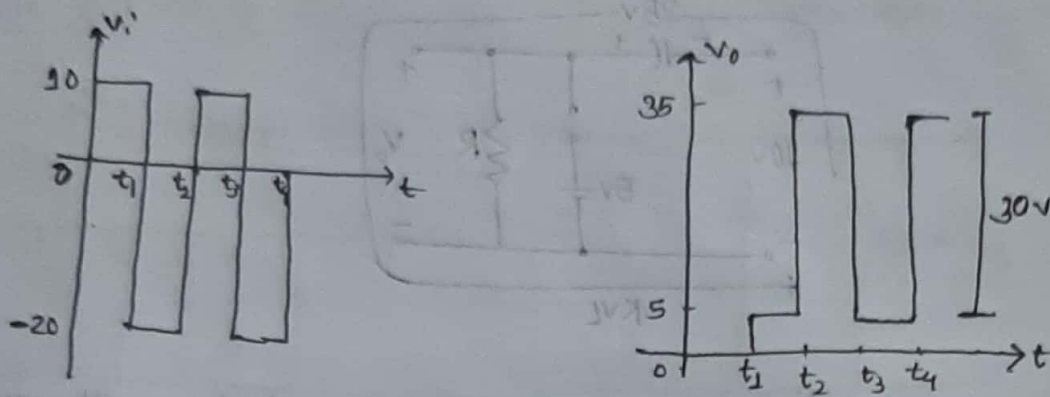
$$= 0.01s = 10ms$$

The total discharge time is therefore,

$$5\tau = 5(10ms) = 50ms$$

During the interval $t_2 \rightarrow t_3$, the capacitor will hold onto its voltage, during the discharge period between pulses of input signal

The resulting output appears as



The output swing of 30V matches the input swing.

Clamper - is an electronic circuit that fixes either the positive or the negative peak excursions of a signal to a defined voltage by adding a variable positive or negative DC voltage to it.

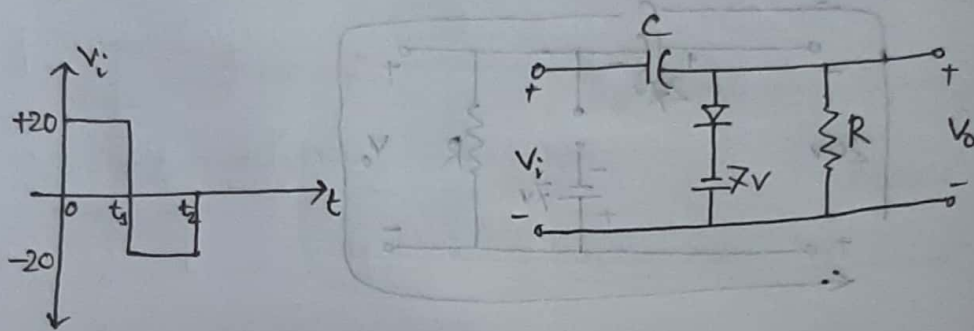
* Application of Clamper circuit:

- used as a voltage multiplier
- used for improving the reverse recovery time.
- used for removing the distortion in the signal
- used as a test equipment.

* Applications of Clipper:

- offer ~~voltage~~ overvoltage protection; therefore it is use for in power supplies for limiting the voltage.
- used for filtering noise in transmitters.
- used in transmitters and receivers of TV
- used for modifying or generating new waveforms such as squares, triangular, etc.

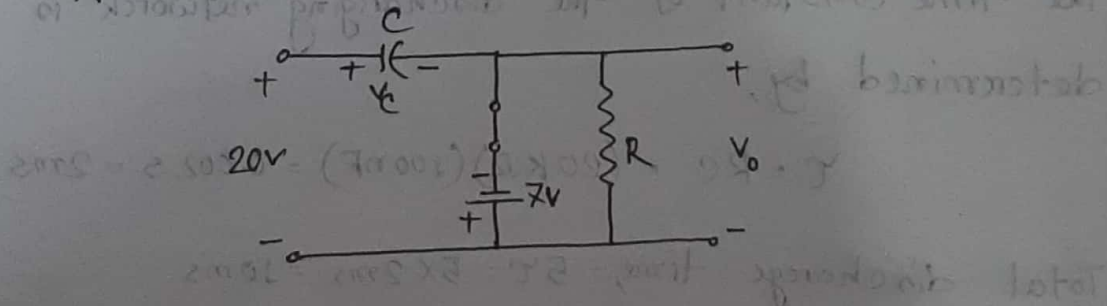
* Analyze output voltage and time constant is analyzed in given clamping network where $f = 2000\text{Hz}$, $R = 20\text{K}\Omega$, $C = 100\text{nF}$



Solution:

Here, $f = 2000\text{Hz}$, $T = \frac{1}{f} = \frac{1}{2000} = 0.0005\text{s} = 0.5\text{ms}$

The analysis will begin with the period $(0 \rightarrow t_1)$ of the input signal since the diode is in short circuit state. The network will appear as:



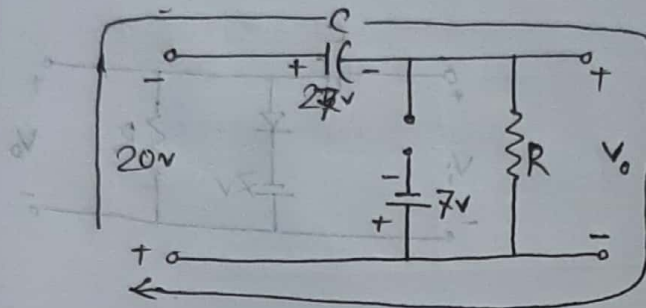
The output is across R but also it is directly across the 7V battery. So, $V_o = 7\text{V}$. Applying KVL around the input loop will result in,

$$+20\text{V} - V_c + 7\text{V} = 0$$

$$\Rightarrow V_c = 27\text{V}$$

The capacitor will charge up to 27V .

For the period $t_1 \rightarrow t_2$, the diode will be in its open circuit state as shown in the following figure-



Applying KVL around the outside loop of the network will result in,

$$+20V + 27V + V_o = 0$$

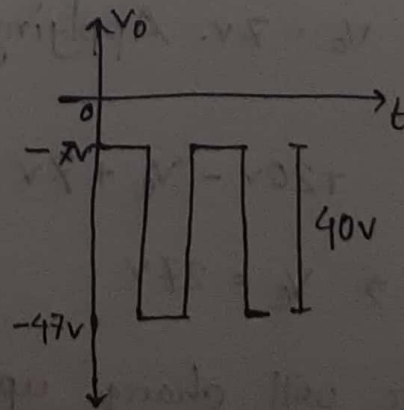
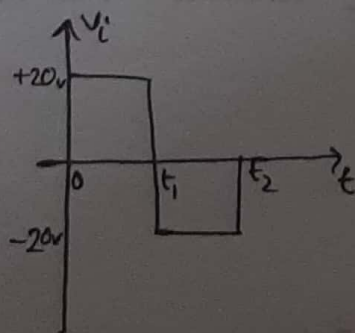
$$\therefore V_o = -47V$$

The time constant of the discharging network is determined by,

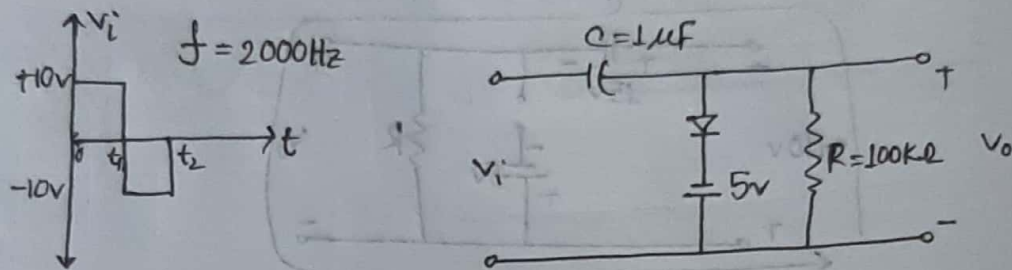
$$\tau = RC = (20K\Omega)(100nF) = 0.002S = 2ms$$

Total discharge time, $5\tau = 5 \times 2ms = 10ms$.

The output signal is,



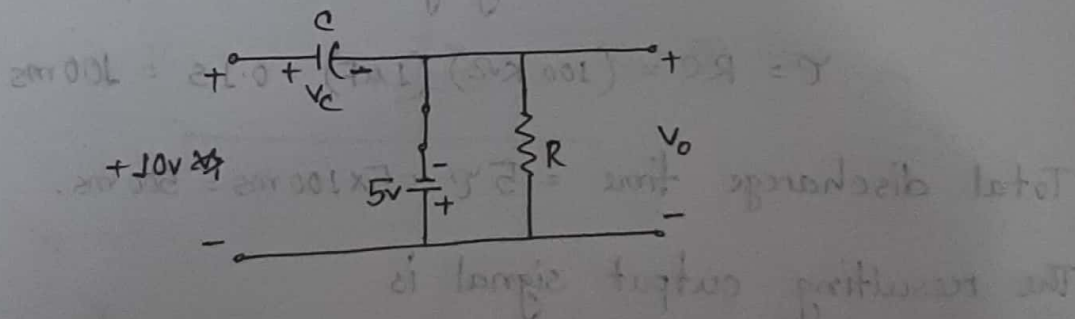
* Analyze output voltage and time constant is analyzed in given clamping network.



Solution:

Here, $f = 2000\text{Hz}$; $T = \frac{1}{f} = \frac{1}{2000} = 0.0005\text{s} = 5\text{ms}$.

The analysis will begin with the period $0 \rightarrow t_1$ as the diode is in its short circuit state. The network will appear as



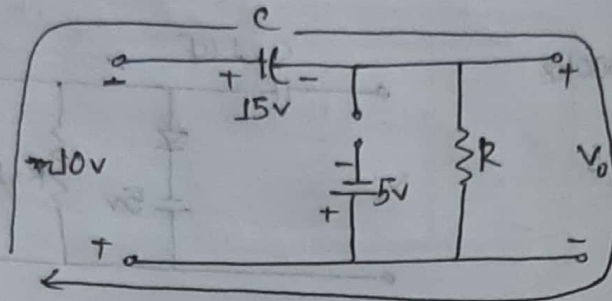
The output voltage is across R and also directly across the 5V battery. So, $V_o = 5\text{V}$. Applying KVL to the input loop of the network, we get,

$$+10\text{V} - V_c + 5\text{V} = 0$$

$$\Rightarrow V_c = 15\text{V}$$

The capacitor will charge upto 15V .

For period $t_1 \rightarrow t_2$, the diode will be in the open circuit state as shown below:



Applying KVL around the outside of the network,

$$+10V + 15V + V_o = 0$$

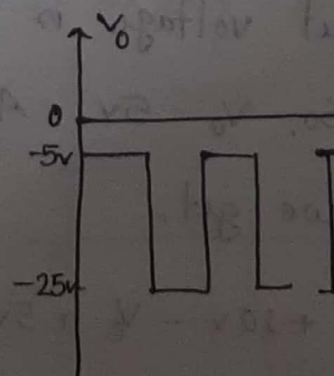
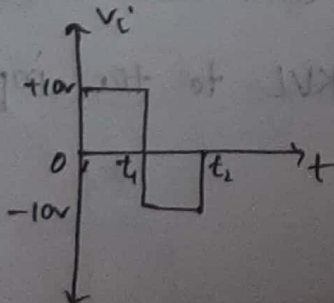
$$\Rightarrow V_o = -25V$$

Time constant of discharging network is,

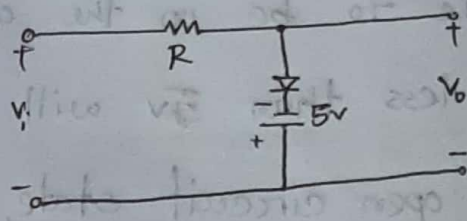
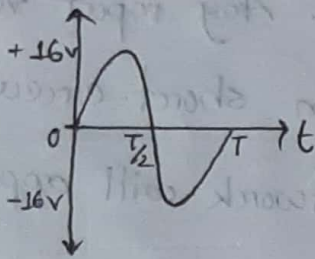
$$\tau = RC = (100 \text{ k}\Omega)(1 \mu\text{F}) = 0.1 \text{ s} = 100 \text{ ms}$$

$$\text{Total discharge time} = 5\tau = 5 \times 100 \text{ ms} = 500 \text{ ms}$$

The resulting output signal is

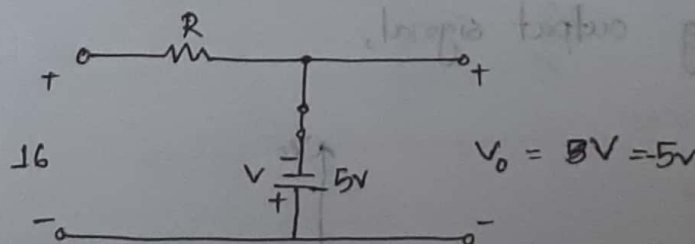


* Determine the output waveform step by step of the given network



Solution:

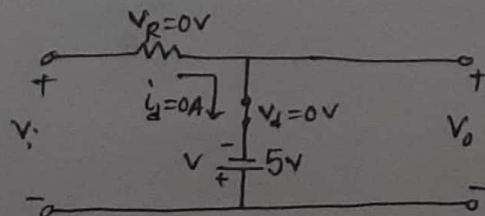
The polarity of dc supply and the direction of the diode suggests that the diode will be in the 'on' state for the positive region of the input signal. The network will appear as



where $V_o = 5V$.

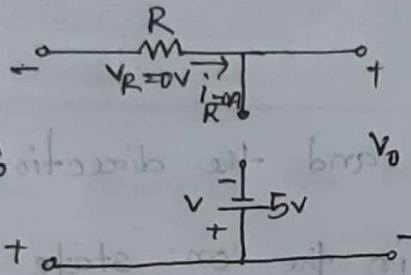
The transition state can be determined by imposing the conditions

$$i_d = 0A, \quad v_d = 0V.$$



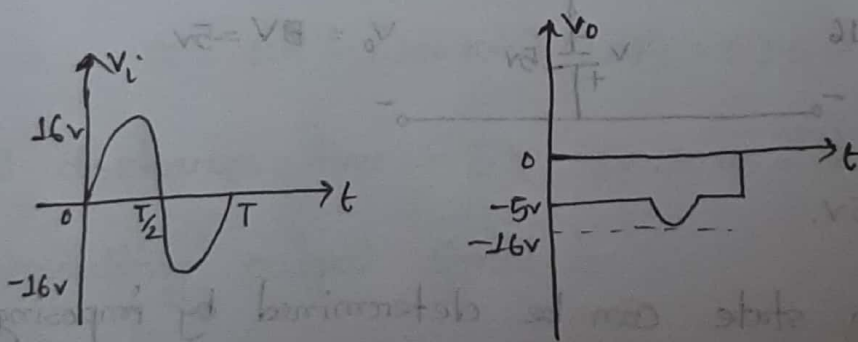
The result is $V_i(\text{transition}) = 5V = 5V$.

The input voltage must be greater than 5V for the diode to be in the off state. Any input voltage less than 5V will result in short-circuited diode. For open circuit state, the network will appear as:

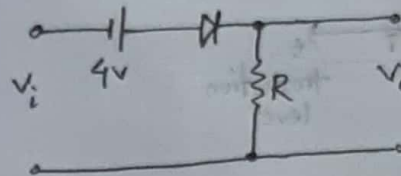
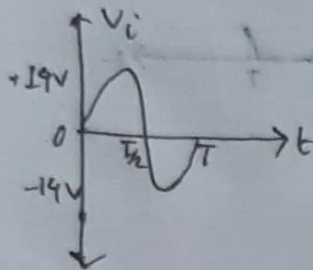


Here, $V_O = V_i = 16V$

The resulting output signal,

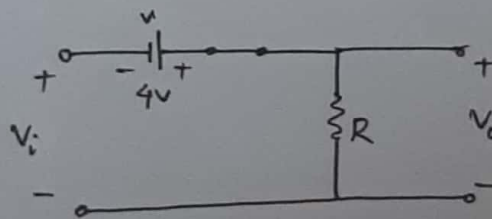


* Analyze the clipper network with given input and determine the output waveform of the given network.



Solution:

The diode will be in the on state for the positive region of the input signal. The network will appear as



$$+V_i + 4V - V_o = 0$$

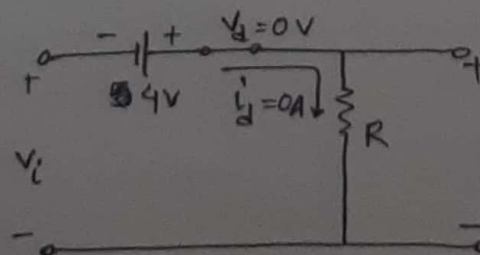
$$V_o = V_i + 4V$$

$$V_o = V_i + 4V$$

$$= 14V + 4V = 18V$$

Substituting $i_d = 0A$, $V_d = 0V$ for the transition levels, we obtain

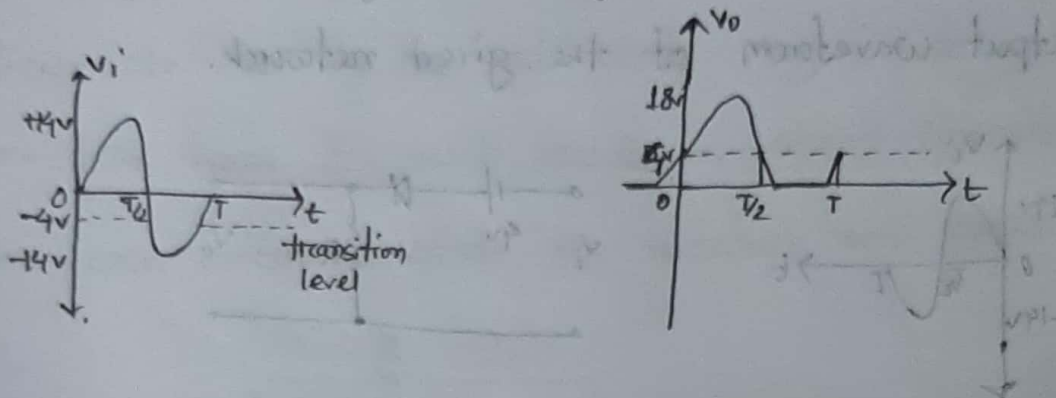
$$V_i = -4V$$



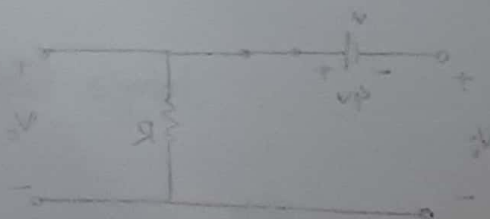
$$V_o = i_R R = i_d R = 0 \cdot R = 0V$$

For V_i more negative than $-5V$ the diode will enter open circuit state. While V_i more positive than $-5V$ will result in short circuit state of the diode.

The input and output voltages will appear as:



The diode will be in the on state for the positive region of the input signal. The output will appear as



$$V_o = V_i + V_D = 14V$$

Substituting $V_D = 0.7V$ for the diode voltage, we get

